The RDBMS implementation on supply chain of a coal company

To Professor Withro Wiggins (IST 659)

Xiaoxuan Tang

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Executive Summary

My research topic highlights on the DBMS implementation on supply chain of a coal company. For the reason of shortage of DBMS, the currency and accuracy of data, under the traditional management environment, is not enough to support SH company's decision-making in strategy layer. Hence, it is definitely required to build a DBMS, as lubrication, to support smooth operation of supply chain.

Facing the highly competitive business environments, building RDBMS is not only the prerequisite of upgrading internal management level, but also the requirement to distinguish business opponents and defeat them.

The DBMS in implementation will support the smooth operation of supply chain (Mentzer, 2001), which consists of coal producing, coal refining, deposit, coal transportation, sales tracking, and payment gathering functions. It not only improves the currency of data processing, but also enhances the accuracy of data result. Simultaneously, it supports to track data from one end of supply chain to the other end, increasing the ability to harness complicated business affairs.

The entity-relationship diagram is provided to promote the deep understanding of DBMS of supply chain. In order to clarify the whole process and keep the project consistency, 12 tables or entities are provided under the support of Microsoft SQL 2003. For the convenience, input forms and output reports are provided under the environment of Microsoft ACESS. Such kinds of distributed DBMSs systems not only simplify the complicacy of data processing, but also improve the efficiency of it.

In order to keep data security on high level and prevent data fraud, some precaution measures must be adopted, such as building firewall to block the unlawful intrusion, obtaining the appropriate database technology, strengthening the aspect of monitoring and controlling of data, separating duties in an appropriate level, controlling access to facilities in the scope of authority allocated, using view technique to hide the valuable data, etc.

Groups of sufficiently descriptive data dictionary are provided. Simultaneously, detailed system catalogs, in describing database objects, are also provided to promote apprehension. The objects involved are tables, forms, reports and queries.

In conclusion, RDBMS on supply chain definitely accelerates the data processing speed, and decreases the workload of managers, simultaneously improves the precision and accuracy of data and information. Most important, it provides a wonderful platform to upgrade

decision-making to strategic level, and make a good preparation in defeating the opponents.

1. Management Overview

1.1. Business problem

My research target focuses on the business problem existed in a large energy company, named SH. The problem is that, on data streaming, the shortage of support from RDBMS severely destroyed the currency and quality of information, simultaneously impeding the normal decision-making process, and threatening the strategy implementation and company survival.

SH is a large energy company focusing on coal mining, coal product dealing, and energy transportation. It mainly provides coal product, a solid state of fuel, to power plants, steel-making plants and chemical plants as power origin or raw materials. Moreover, its energy transportation service is highly efficient and highly competitive.

In usual, business decisions are made on the basis of information and materials passed from low level to high layer. However, under the situation of lacking RDBMS support, the currency and quality of information, which can not be guaranteed, impede the decision-making process. At the same time, without strong support coming from databases, it is very difficult to promptly obtain and process the information coming from competitors. Further more, it is hard to improve the efficiency of company's supply chain operation without introducing the database management platform. Hence, it is definitely necessary to use database and RDBMS, as a bridge, to solve the information problem faced by SH company.

Under Porter's five force theory, combining the competitive environments SH company faced, there are five key business decisions can not be made without having access to the right data. They are decision-making about competitive rivalry, supplier power, buyer power, barrier to entry, and the threat of substitutes (Porter, 1980). The decision-making in such five aspects depends on tons of information collected from all of directions. How to solve these problems is the key of business success of a company.

1.2. Project scope

My project will put emphasis on one aspect of competitive intelligence (Fleisher, 2003), which is aiming at increasing the efficiency on coal supply chain of SH Company, by introducing database administration.

The functions of supply chain on SH, I prepare to support, include coal producing, refining, deposit, transportation, sales tracking, and

payment gathering functions. However, there are still some functions outside supply chain not included, such as accounting module, equipment management module, and human resources management module. In future, it maybe appears a wonderful way to combine supply chain systems with these out-layer modules.

1.3. Solution

The new database and DBMS change the way work gets done, on subsequent aspects.

The first, it makes the operation of supply chain more frequent and smooth than before. The previous data handling method by hand, not only requires employing many people, but also easily makes mistakes. Nowadays, new DBMS implementation not only decreases the employee numbers, but also makes the data flow smoothly. The second, it decreases errors and inharmony in data. The previous data handling method are easy to make mistakes. Nowadays, it improves the precision by implementing integrity check in RDBMS. The third, it provides much more current data than before. The previous data handling method requires so long time to process data that it usually delays decision-making. Nowadays, it provides strong data support in a definitely faster speed, very helpful in grasping fleeting business opportunities. The fourth, it provides a comprehensive and integral solution for decision-making by handling tons of valuable data. The previous data handling method merely provide dispersed and distracted data, not enough to support high-level decision-making in time. Nowadays, it can provide a bunch of data set to support decision-making in different layers, whatever lower level or higher ones. The fifth, it supports tracking data from one end of supply chain to the other end. The previous data handling method is completely impossible to track data on long chain in high accuracy. Nowadays, it can quickly track them in the scope of whole supply chains or any part of it.

Except the descriptions above, there are still some specific examples of business decisions, which should be clarified. The first, by introducing RDBMS, the companies can track the residual quantity of coal reserves at any time. Hence, it provides a very helpful precaution to the decision-making layer, which should make a decision whether to purchase some new mines to keep reserves on a suitable level. The second, the companies can track which kinds of coal refining technique can produce biggest benefit. It provides great help on company’s decision-making in choosing most suitable refining technique for coal product. The third, the companies can track the relationship between patron’s payments speed and the speed of transportation provided by companies. It is definitely helpful to help companies put emphasis on improve supply chain

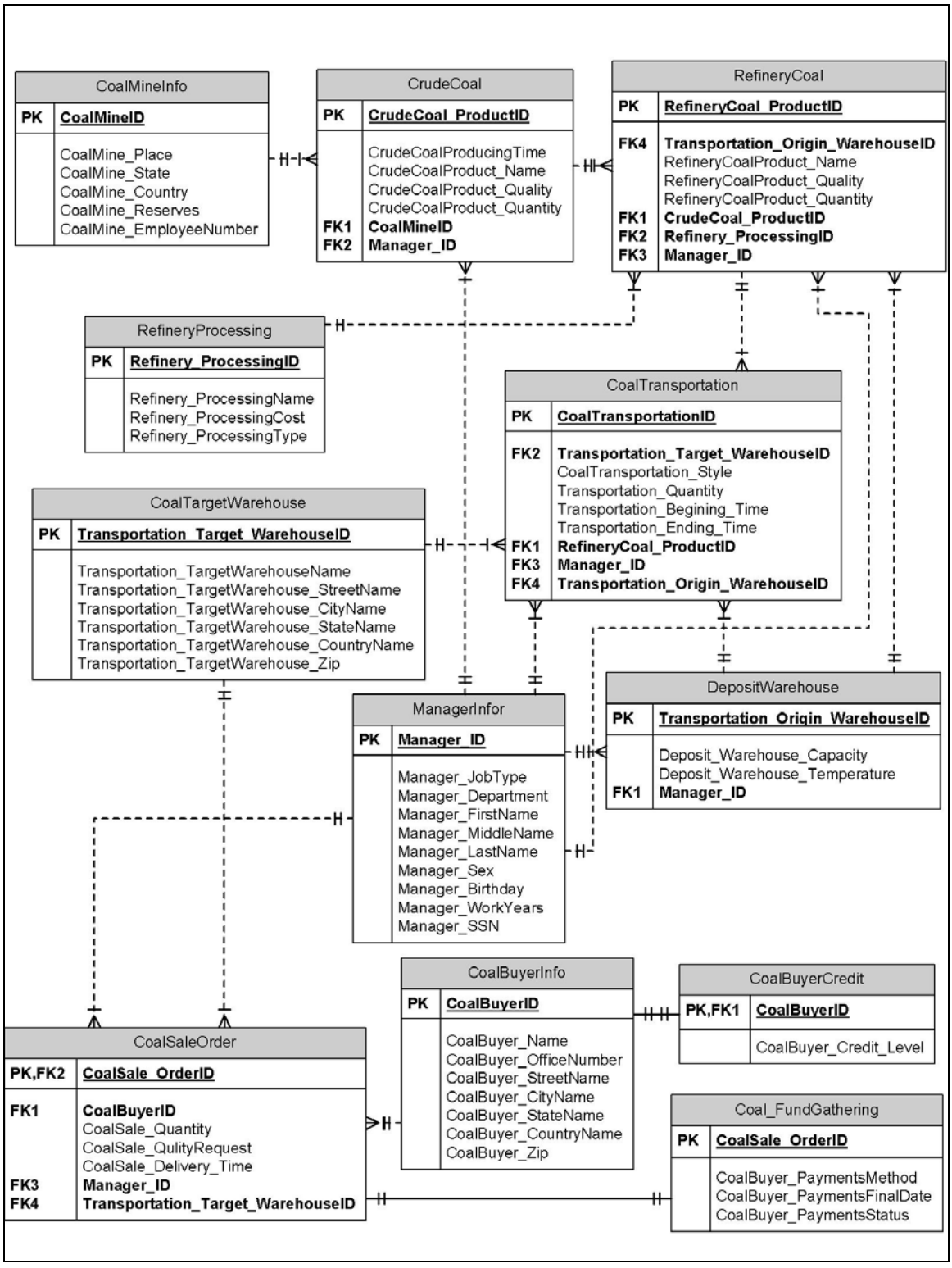
efficiency. The fourth, the companies can track which kinds of refining coal are most popular in market, and simultaneously find which coal mine produces such kinds of coals. In this way, the companies can adjust the coal product structure among different coal mines. The fifth, the companies can track the low-quality coal product. As a result, it can track the related manager with responsibility.

1.4. Assumptions & constraints

I make three assumptions. The first, the accounting module, employee management module (not include managers), and equipment module, are outside of supply chain DBMS system. The second, the deposit warehouse is the origin or starting point of coal transportation. Moreover, the deposit warehouses belong to coal companies and near the coalmine. The third, the target warehouses should locate in patrons' side and belong to buyers.

During the process of building RDBMS, the only obstacle, I encounter, is how to clarify and connect different entity which is dispersed and located in the supply chain. Finally I find a wonderful way which employs transportation entity as a bridge to connect coal production and sales ones. Simultaneously, I use manager entity as a second bridge to connect most of other entities.

1.5. Database model



1.6. Business rules

1.6.1. CoalBuyerInfo

Different coal buyers have different name and address. It is not allowed to have completely same characteristics for two different buyers.

1.6.2. Coal_FundGathering

During the process of fund gathering, only two paying method are permitted which are '1' equal to 'Draft', and '2' equal to 'Check'. There are merely two payments status allowed, which are '1' equal to 'Clear payments', '2' equal to 'Not clear payments'. One sales order can only have one payments method, and one payment status.

1.6.3. CoalBuyerCredit

Every coal buyer must have the credit level evaluation. Without it, sale department can not sign an order or agreement with buyers.

1.6.4. ManagerInfor

One manager should not work as employee of two departments at the same time.

1.6.5. CoalTargetwarehouse

It is not allowed that two target warehouses are totally same in all aspects. One target warehouse can accept coal products from many origin warehouses. One target warehouse can store coal products which belong to many sale orders

1.6.6. CoalSaleOrder

The one buyer can sign many different sale order, with the time passing, to satisfy the seasonal energy requirments. Every coal sale order can only have one buyer. Every sale order can only have one payment method and one payment status.

1.6.7. CoalMineInfo

It is not allowed that two coal mines are totally same in all aspects. One coal mine can produce many batch of coal products, with the time passing.

1.6.8. CrudeCoal

One batch of crude coal product, which is differentiated by "Crudecoal_productID", can only be produced by one coal mine. One batch of crude coal product can be finished processing in different times by rifinery factory, according to the processing capacity and order request.

1.6.9. RefineryProcessing

It is not allowed that two refinery processing techniques have totally same characteristics in all aspects. One refinery processing technique can be used to process many batch of coal products.

1.6.10. DepositWarehouse

One deposit warehouse can store many batch of coal products coming from refinery factory. Deposit warehouse are located near the coal mine and refinery factory, so they have no detailed address.

1.6.11. RefineryCoal

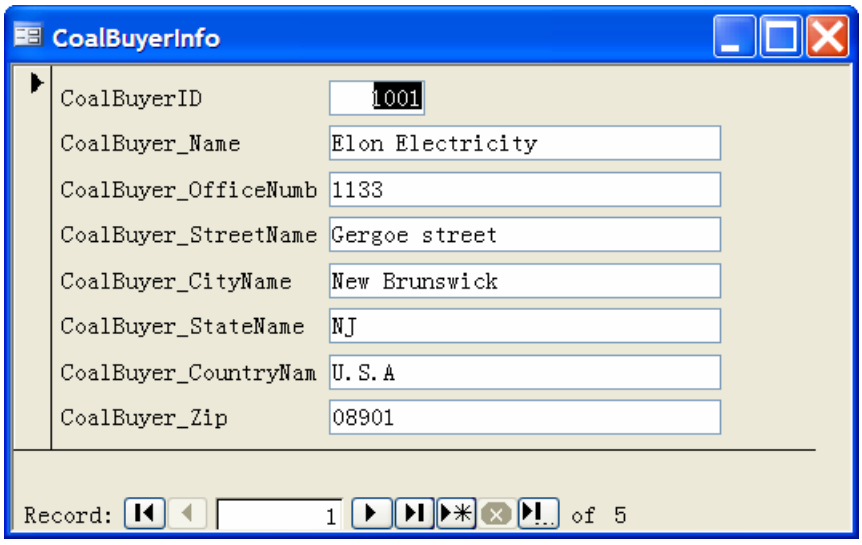
One batch of refinery coal can only come from one batch of crude coal, simutaneously come from only one coal mine. One batch of refinery coal can only employ one refinery processing technique. One batch of refinery coal can only go into one deposit warehouse.

1.6.12. CoalTransportation

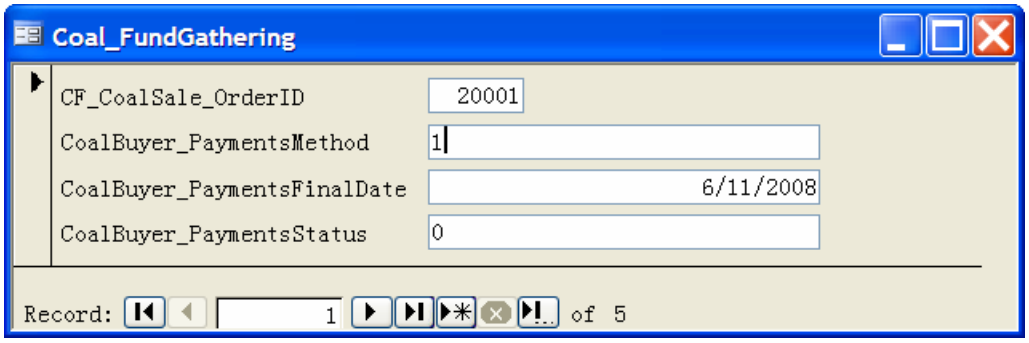
One transportation task, which is differentiated by CoalTransportationID, can only carry and transport one batch of refinery coal from Deposit warehouse (origin warehouse) to target warehouse. The coal products in deposit warehouse (origin warehouse) must be transported to target warehouse which belongs to buyers.

1.7. Input forms

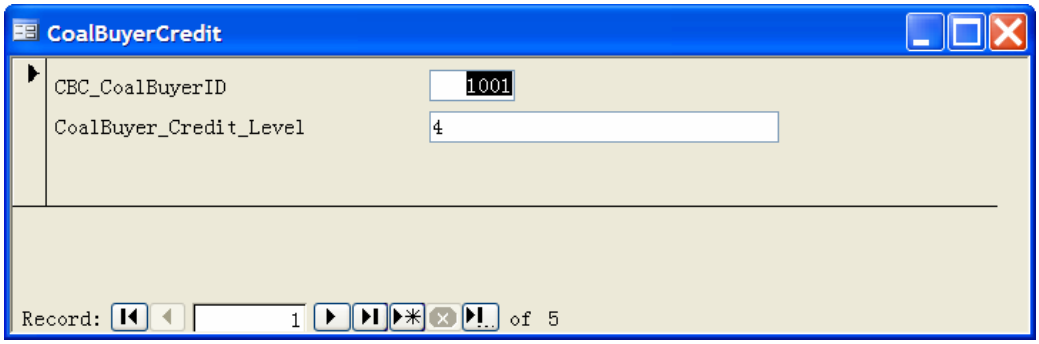
1.7.1. Form_CoalBuyerInfo



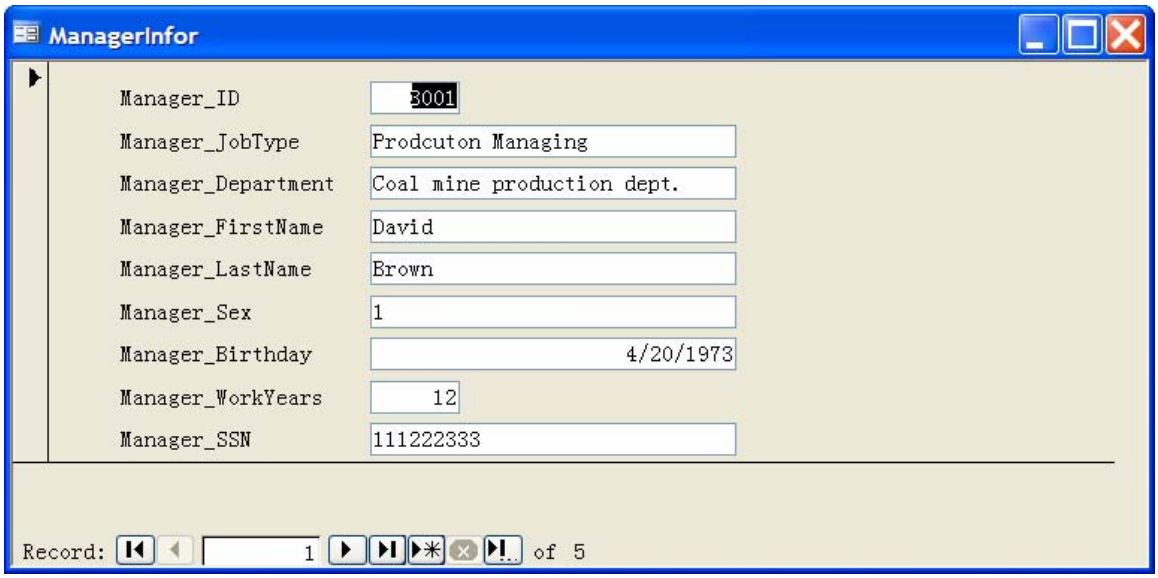
1.7.2. Form_Coal_FundGathering



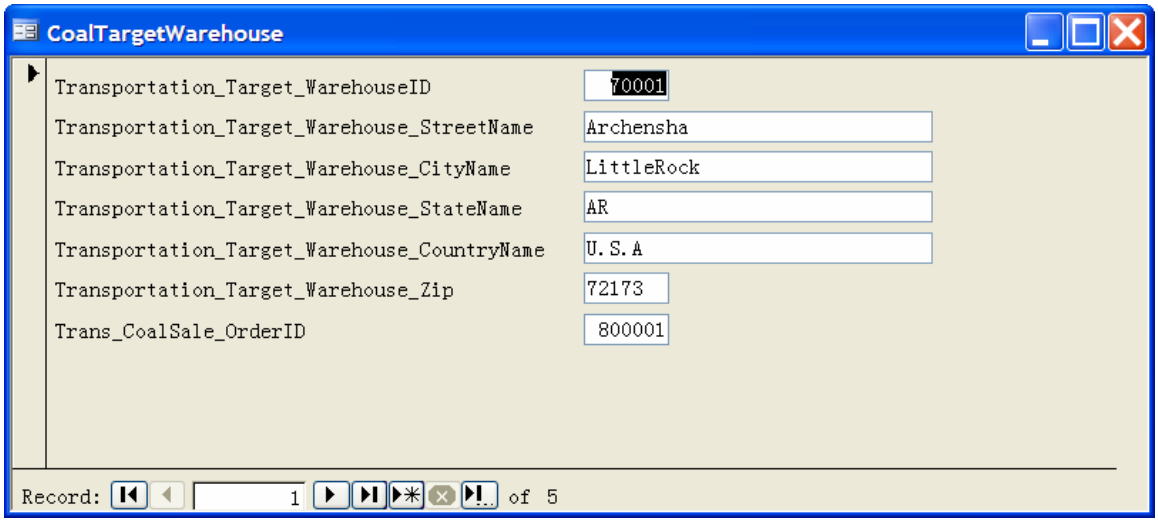
1.7.3. Form_CoalBuyerCredit



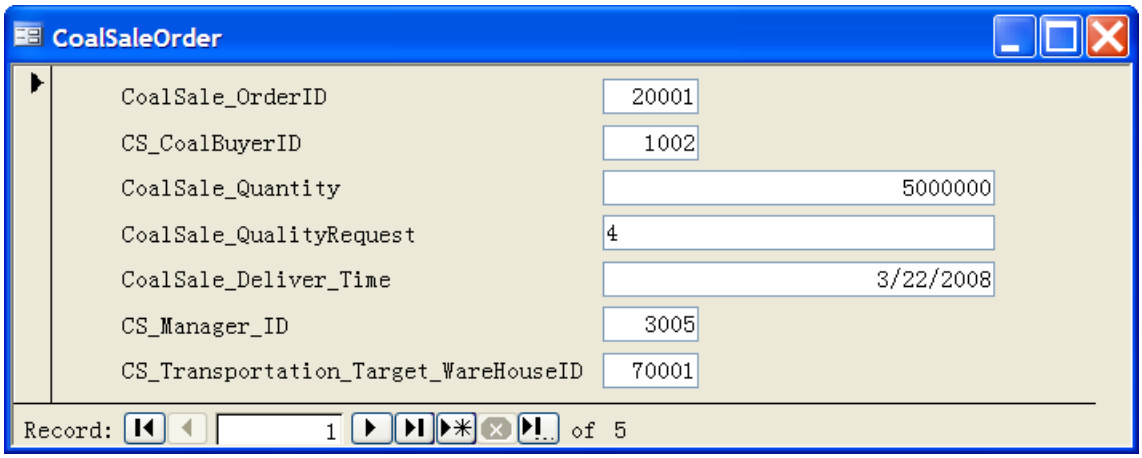
1.7.4. Form_ManagerInfor



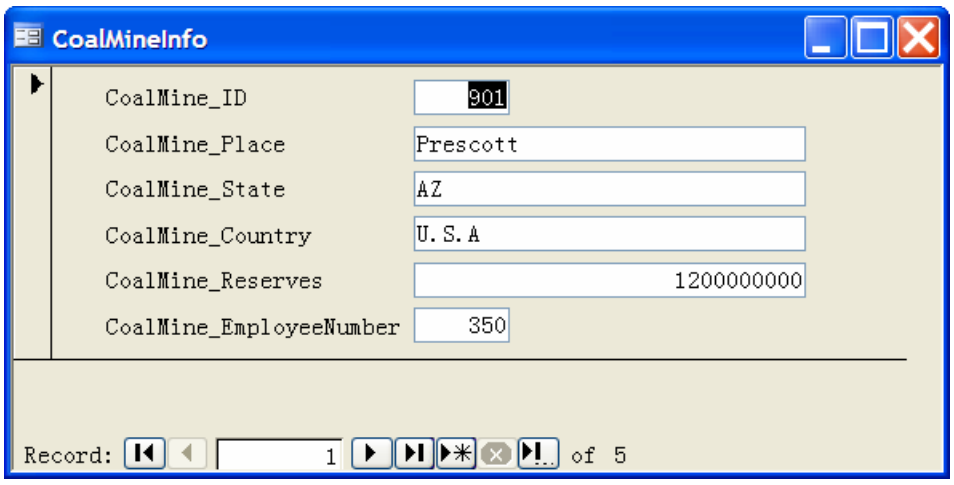
1.7.5. Form_CoalTargetWarehouse



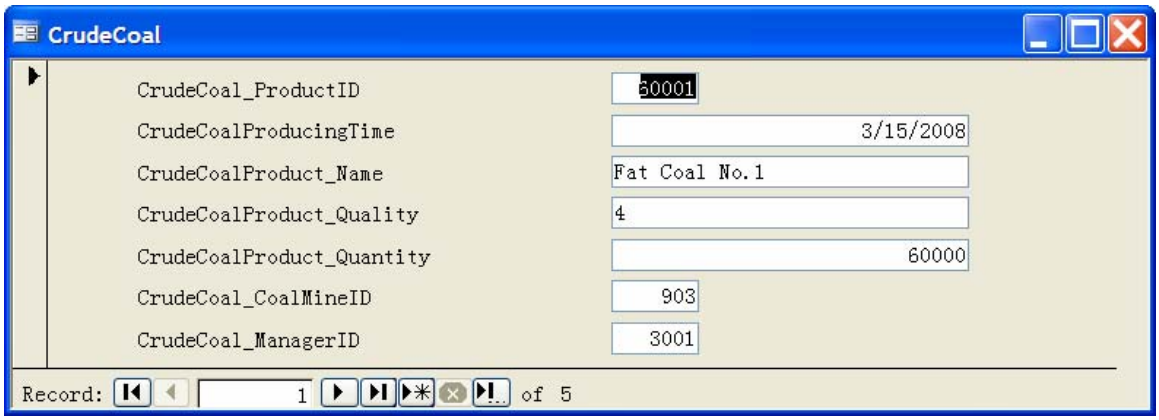
1.7.6. Form_CoalSaleOrder



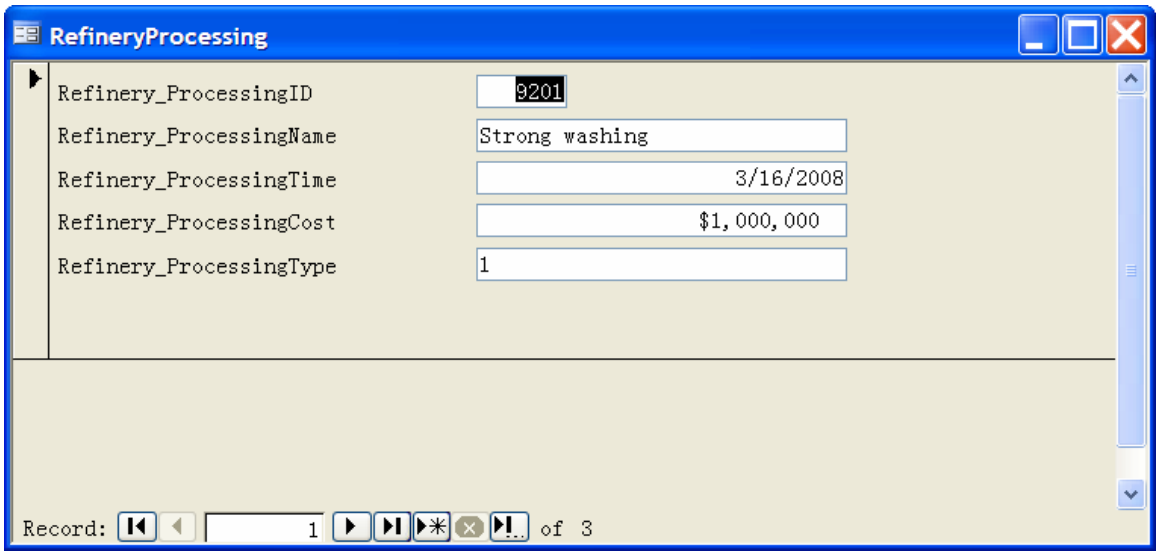
1.7.7. Form_CoalMineInfo



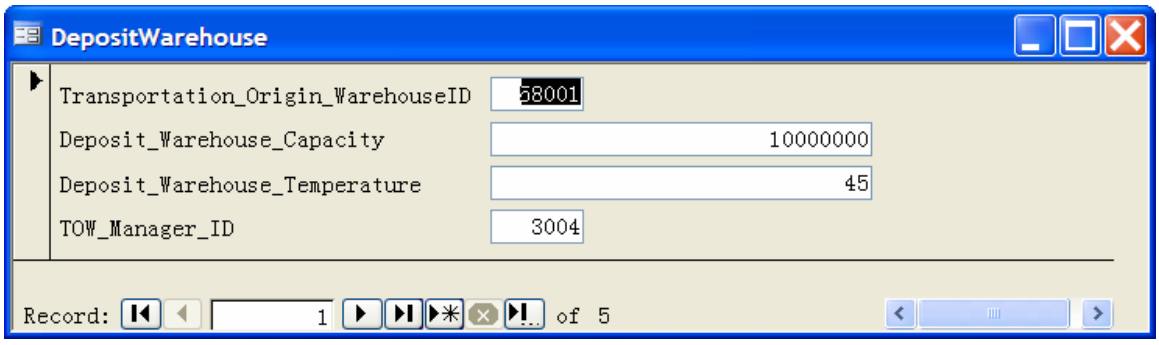
1.7.8. Form_CrudeCoal



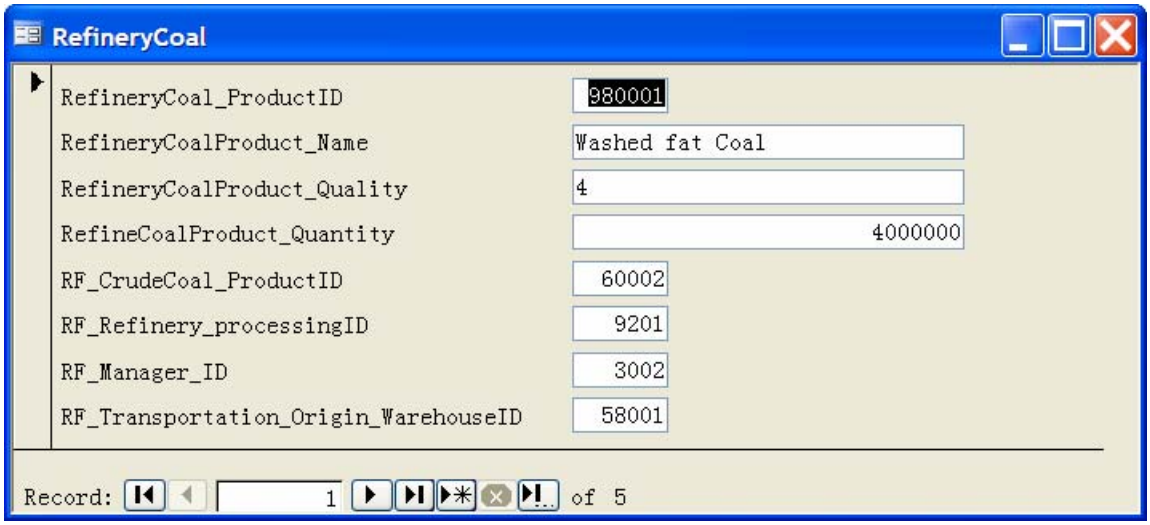
1.7.9. Form_RefineryProcessing



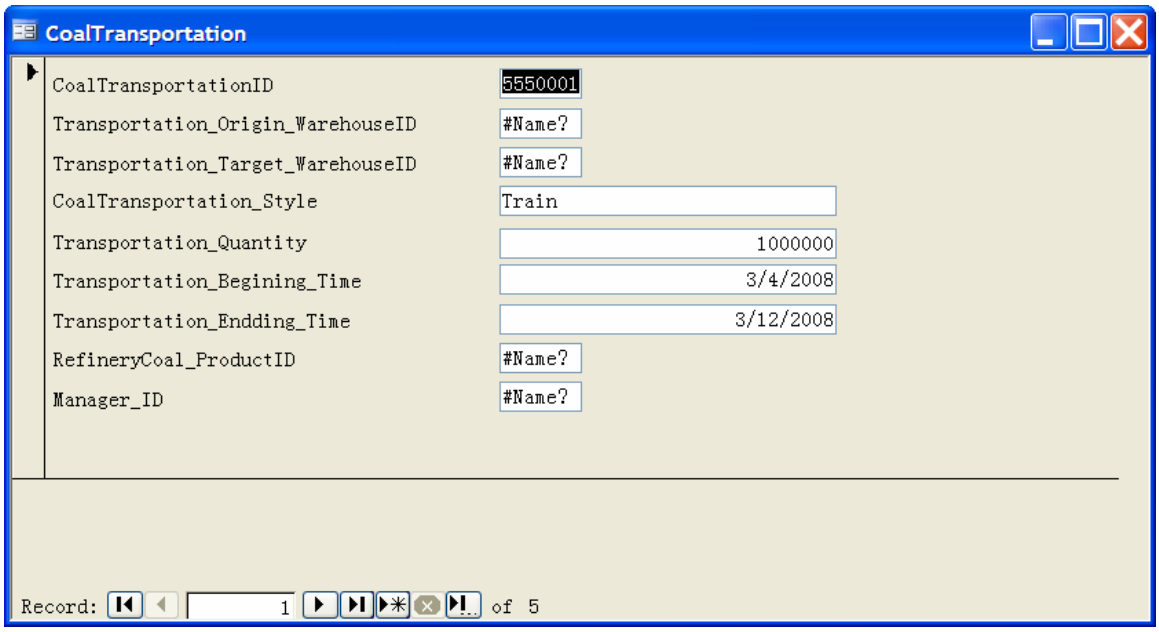
1.7.10. Form_DepositWarehouse



1.7.11. Form_RefineryCoal

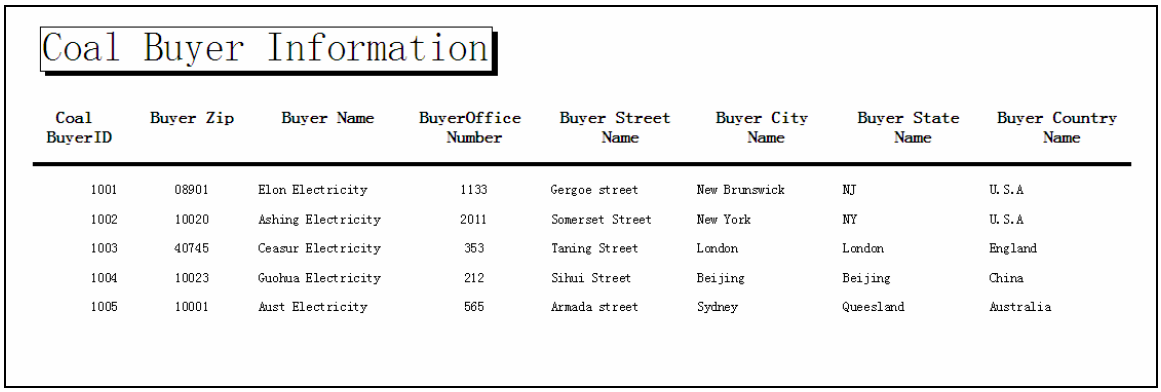


1.7.12. Form_CoalTransportation



1.8. Output Reports

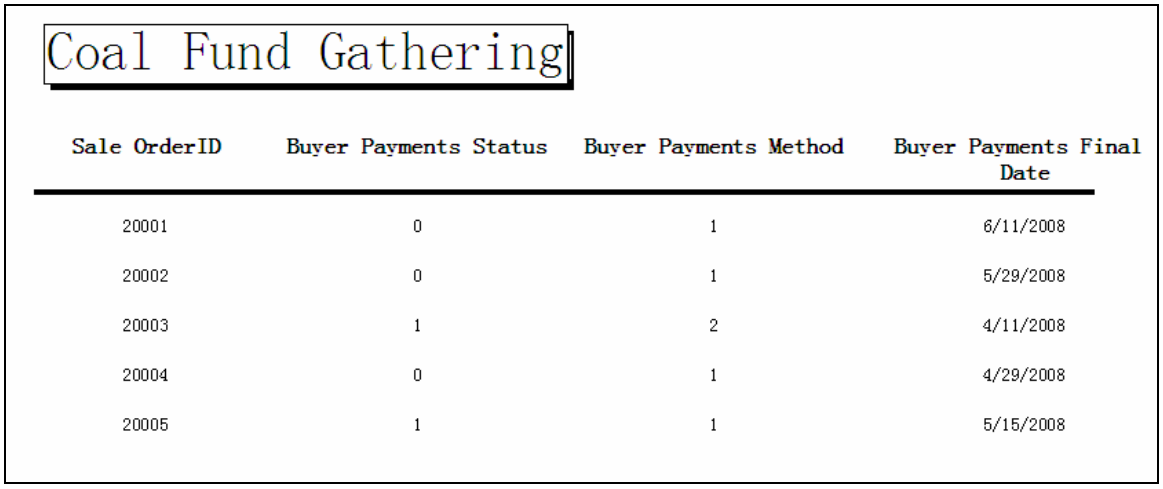
1.8.1. Rpt_CoalBuyerInfo



Coal Buyer Information

Coal BuyerID	Buyer Zip	Buyer Name	BuyerOffice Number	Buyer Street Name	Buyer City Name	Buyer State Name	Buyer Country Name
1001	08901	Elon Electricity	1133	Gergoe street	New Brunswick	NJ	U.S.A
1002	10020	Ashing Electricity	2011	Somerset Street	New York	NY	U.S.A
1003	40745	Ceasur Electricity	353	Taning Street	London	London	England
1004	10023	Guohua Electricity	212	Sihui Street	Beijing	Beijing	China
1005	10001	Aust Electricity	565	Armada street	Sydney	Queesland	Australia

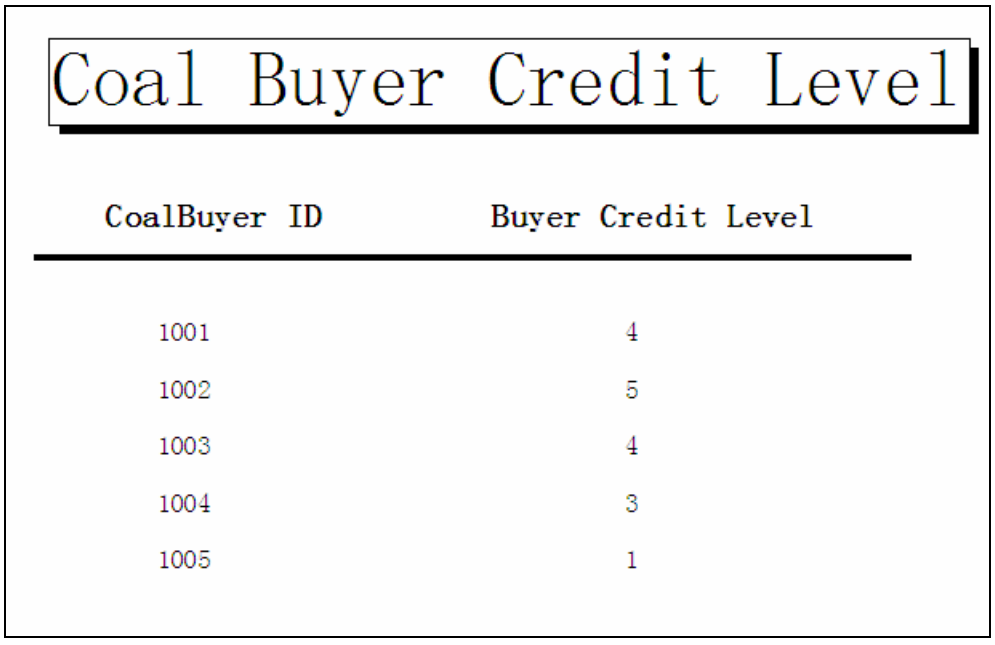
1.8.2. Rpt_Coal_FundGathering



Coal Fund Gathering

Sale OrderID	Buyer Payments Status	Buyer Payments Method	Buyer Payments Final Date
20001	0	1	6/11/2008
20002	0	1	5/29/2008
20003	1	2	4/11/2008
20004	0	1	4/29/2008
20005	1	1	5/15/2008

1.8.3. Rpt_CoalBuyerCredit



Coal Buyer Credit Level

CoalBuyer ID	Buyer Credit Level
1001	4
1002	5
1003	4
1004	3
1005	1

1.8.4. Rpt_ManagerInfor

Manager Information

Manager ID	Manager Department	Manager Last Name	Manager Birthday	Manager Job Type	Manager First Name	Manager Gender	Manager Work Years	Manager SSN
3001	Coal mine production dept.	Brown	4/20/1973	Prodcuton Managing	David	1	12	111222333
3002	Refinery factory	Davidson	7/20/1976	Refinery managing	Michelle	1	9	123123456
3003	Transportation dept.	Jane	8/20/1979	Transportation managing	Mary	0	7	133555666
3004	Warehouse dept.	Lee	9/20/1980	Warehouse managing	Peter	1	8	333000666
3005	Sales Dept.	Moly	9/21/1981	Sales managing	Rebecca	0	7	888999000

1.8.5. Rpt_CoalTargetWarehouse

Coal Target Warehouse Information

Target WarehouseID	Target Warehouse Street Name	Target Warehouse City Name	Target Warehouse State Name	Target Warehouse Country Name	Target Warehouse Zip	Sale Order ID
70001	Archensha	LittleRock	AR	U.S.A	72173	800001
70002	Fuda	Cookeville	TN	U.S.A	38501	800002
70003	Yida	Tuscollosa	AL	U.S.A	35401	800003
70004	Boda	Galveston	TX	U.S.A	77550	800004
70005	Bernard	Oxnard	CA	U.S.A	93030	800005

1.8.6. Rpt_CoalSaleOrder

Coal Sale Order Information

Sale Order ID	Buyer ID	Sale Quantity	Sale Quality Request	Sale Deliver Time	Manager ID	Transportation Target WareHouseID
20001	1002	5000000	4	3/22/2008	3005	70001
20002	1004	6000000	3	3/29/2008	3005	70002
20003	1003	5600000	5	4/19/2008	3005	70004
20004	1005	10000000	4	4/11/2008	3005	70005
20005	1001	3000000	6	5/1/2008	3005	70003

1.8.7. Rpt_CoalMineInfo

Coal Mine Information

Coal MineID	CoalMine Reserves	CoalMine Employee Number	CoalMine Place	CoalMine State	CoalMine Country
901	1200000000	350	Prescott	AZ	U.S.A
902	700000000	900	Evanston	WY	U.S.A
903	500000000	600	Hutchinson	KS	U.S.A
904	800000000	460	Cody	WY	U.S.A
905	900000000	500	Columbus	IN	U.S.A

1.8.8. Rpt_CrudeCoal

Crude Coal Information

Crude Coal ProductID	Crude Coal CoalMine ID	CrudeCoal Producing Time	CrudeCoal Product Name	CrudeCoal Product Quality	CrudeCoal Product Quantity	CrudeCoal Manager ID
60001	903	3/15/2008	Fat Coal No.1	4	60000	3001
60002	901	4/1/2008	Fat Coal No.5	3	70000	3001
60003	905	3/27/2008	Gas Coal No.3	4	80000	3001
60004	902	3/31/2008	Forge Coal No.3	6	50000	3001
60005	904	3/30/2008	Lignite No.1	5	98000	3001

1.8.9. Rpt_RefineryProcessing

Refinery Processing Information

Refinery Processing ID	Refinery Processing Name	Refinery Processing Cost	Refinery Processing Type
9201	Strong washing	$1,500,000	1
9202	Middle washing	$1,300,000	2
9203	Light washing	$800,000	3

1.8.10. Rpt_DepositWarehouse

Deposit Warehouse Informatio

WarehouseID	Warehouse Capacity	Warehouse Temperature	Manager ID
58001	10000000	45	3004
58002	20000000	50	3004
58003	15000000	50	3004
58004	18000000	45	3004
58005	16000000	50	3004

1.8.11. Rpt_RefineryCoal

Refinery Coal Information

Refinery Coal ProductID	Refinery Coal Product Quality	Refinery processing ID	Refinery Coal Product Name	Refinery Coal Product Quantity	Crude Coal Product ID	Manager ID	Origin Warehouse ID
980001	4	9201	Washed fat Coal	4000000	60002	3002	58001
980002	5	9202	Washed fat coal	5000000	60002	3002	58001
980003	3	9202	Washed Lignite	3000000	60005	3002	58002
980004	4	9203	Washed gas coal	2000000	60003	3002	58003
980005	5	9201	Washed forge coal	1000000	60004	3002	58004

1.8.12. Rpt_CoalTransportation

Coal Transportation Informatio

Transportation ID	Origin WarehouseID	Target WarehouseID	Transportation Style	Transportation Quantity	Transportation Begining Time	Transportation Endding Time	Refinery Coal Product ID	Manager ID
5550001	58001	70001	Train	1000000	3/4/2008	3/12/2008	980002	3003
5550002	58002	70001	Train	2000000	3/6/2008	3/18/2008	980003	3003
5550003	58003	70005	Ship	1500000	4/3/2008	4/12/2008	980004	3003
5550004	58004	70002	Train	1700000	4/5/2008	4/15/2008	980005	3003
5550005	58003	70004	Train	1300000	2/6/2008	2/16/2008	980004	3003

1.9. Support requirements

- Database backup/recovery

  There are three methods to ensure database backup. The first, database administrator must backup database periodically by using full backup method, consisting of whatever hot backup or cold backup. The second, database administrator must make checkpoint to ensure the partial data safety. The third, the hardware administrator should adopt raid 0 or raid 1 to guarantee the data safety.

- Backup file retention

Database administrator must guarantee the backup file retained in credible media, such as DVD discs or tapes.

- Data audit control

  There are many ways to strengthen the audit control. The first, on the aspect of hardware, it is best to purchase the computers with the functions of preventing electronic signal emission. The second, to strengthen access control is a good way in isolation of unrelated person. The third, DBA should adopt integrity control. The fourth, DBA should use database logs to detect and track unlawful intruders. The fifth, DBA should frequently use “View” to hide real data. The sixth, DBA can add some encryption middleware to control safety. The seventh, DBA should build a system to control and monitor the DBMS input and output, memory, and CPU using status.

- Data Security and privacy

  There are many ways to strengthen the data security and protect privacy. The first, DBA should make a data risk assessment. The second, DBA should analyze the data threats, risks and simultaneously prepare some precaution methods. The third, it is important on purchasing some hardware to ensure database has a solid and strong architecture. The fourth, DBA should introduce some new technology to improve the safety of data. The fifth, DBA must strengthen access monitor and control. The sixth, some responsibilities and duties had better be allocated to different employees for the sake of increasing the mutual audit. The seventh, DBA should require and guarantee every employee changing their password in a fixed time interval.

2. Supporting Documentation

2.1. Data dictionary

Table name: CoalBuyerInfo

Description: The CoalBuyerInfo table depicts detailed demographic information regarding buyers whose identity are companies in usual.

Name	Desc	Type/ Size	For mat	Req'd	Domain	Default/ Derived	Key	Referenced Table
CoalBuyerID	Coal buyer ID	Int		Y	0001-9999		PK	
CoalBuyer_Name	Coal buyer Name	Varchar(30)		Y				
CoalBuyer_OfficeNumber	Coal buyer office room number	Varchar(10)		Y				
CoalBuyer_StreetName CoalBuyer_CityName	Coal buyer street name	Varchar(30)		Y				
CoalBuyer_StateName	Coal buyer state name	Varchar(30)		Y				
CoalBuyer_CountryName	Coal buyer country name	Varchar(30)		Y				
CoalBuyer_Zip	Coal buyer zip	Varchar(10)		Y				

Table name: Coal_FundGathering

Description: The Coal_FundGathering table mainly depicts whether the fund, relating with sale order, has been gathered or received.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
CF_CoalSale_OrderID	Coal sales order ID	Int		Y	00001-99999		PK	
CoalBuyer_PaymentsMethod	The payment method	Varchar(10)		Y				
CoalBuyer_PaymentsFinalDate	The final date of payments	Datetime(8)	Mm/dd/yy	Y				
CoalBuyer_PaymentsStatus	The payment status	Varchar(10)		Y	1,0			

Table name: CoalBuyerCredit

Description: The CoalBuyerCredit table mainly depicts the credit level of coal buyers. It is viewed as a warrant or foundation of payments.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
CBC_CoalBuyerID	Coal buyers ID	Int		Y	0001-9999		PK	
CoalBuyer_Credit_Level	The credit level of coal buyers	Varchar(10)		Y	1-6			

Table name: ManagerInfor

Description: The ManagerInfor table mainly depicts the demographic information of all managers who have great responsibilities on the supply chain

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
Manager_ID	Manager ID	int		Y	0001-9999		PK	
Manager_JobType	Job type	varchar(40)		Y				
Manager_Department	department	varchar(40)		Y				
Manager_FirstName	First name	varchar(30)		Y				
Manager_MiddleName	Middle name	varchar(20)		Y				
Manager_LastName	Last name	varchar(30)		Y				
Manager_Sex	Gender	varchar(1)		Y	0,1			
Manager_Birthday	Birthday	Datetime	Mm/dd/yy	Y				
Manager_WorkYears	Work years	int		Y				
Manager_SSN	SSN	varchar(9)		Y				
				Y				
				Y				

Table name: CoalTargetWarehouse

Description: The table of CoalTargetWarehouse mainly depicts the ware house information of the buyers. This warehouse usually is the coal transportation target which is used to store the coal product sold to buyers. These kinds of warehouse usually belong to users or patrons.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
Transportation_Target_WarehouseID	Target warehouse ID of transporation	int		Y	00001-99999		PK	
Transportation_Target_WarehouseName	Target warehouse name	Varchar(30)		Y				
Transportation_Target_Warehouse_StreetName	Target warehouse street name	Varchar(50)		Y				
Transportation_Target_Warehouse_CityName	Target warehouse city name	Varchar(30)		Y				
Transportation_Target_Warehouse_StateName	Target warehouse state name	Varchar(10)		Y				
Transportation_Target_Warehouse_CountryName	Target warehouse country name	Varchar(30)		Y				
Transportation_Target_Warehouse_Zip	Target warehouse zip	Varchar(10)		Y				
Trans_CoalSale_OrderID	Sales order ID of coal	int		Y	000001-999999		FK	CoalSaleOrder

Table name: CoalSaleOrder

Description: The CoalSaleOrder table depicts the detailed information of sale orders, highlight on who the buyer is, what the buyer's requirement for coal products is, and where buyers want the coal to be transported.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
CoalSale_OrderID	The coal sales order ID	Int		Y	00001-99999		PK/FK	Coal_FundGathering
CS_CoalBuyerID	Coal buyer ID	Int		Y	0001-9999		FK	CoalBuyerInfo
CoalSale_Quantity	The quantity of coal sales	Decimal(14,2)		Y				
CoalSale_QualityRequest	The requested coal quality	varchar(10)		Y				
CoalSale_Deliver_Time	The delivery time on sale order	Datetime(8)	Mm/dd/yy	Y				
CS_Manager_ID	Manager ID	int		Y	0001-9999		FK	ManagerInfor
CS_Transportation_Target_WareHouseID	The target warehouse of transportation	Int		Y	00001-99999		FK	CoalTargetWarehouse

Table name: CoalMineInfo

Description: The CoalMineInfo table contains basic information description regarding coal mines.

Name	Desc	Type/ Size	For mat	Req'd	Domain	Default/ Derived	Key	Referenced Table
CoalMine_ID	Coal mine ID	Int		Y	001-999		PK	
CoalMine_Place	Coal mine city	Varchar(60)		Y				
CoalMine_State	Coal mine state	Varchar(20)		Y				
CoalMine_Country	Coal mine country	Varchar(30)		Y				
CoalMine_Reserves	Coal mine reserves	Decimal(15,2)		Y				
CoalMine_EmployeeNumber	The total employee number of coal mine	Int		Y	0001-9999			

Table name: CrudeCoal

Description: The table of CrudeCoal constrains the information related with crude coal production.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
CrudeCoal_ProductID	The product ID of crude coal	int		Y	00001-99999		PK	
CrudeCoalProducingTime	The producing time of crude coal	datetime	Mm/dd/yy	Y				
CrudeCoalProduct_Name	Crude coal product name	varchar(50)		Y				
CrudeCoalProduct_Quality	Crude coal product quality	varchar(30)		Y				
CrudeCoalProduct_Quantity	Crude coal product quantity	decimal(12, 2)		Y				
CrudeCoal_CoalMineID	Coal mine ID	int		Y	001-999		FK	CoalMineInfo
CrudeCoal_ManagerID	Manager ID who handle the crude coal production	int		Y	0001-9999		FK	ManagerInfor

Table name: RefineryProcessing

Description: The table of RefineryProcessing mainly depicts the information of refining process which includes processing type, time, and cost per unit.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
Refinery_ProcessingID	Refinery processing ID	int		Y	0001-9999		PK	
Refinery_ProcessingName	Refinery processing name	varchar(50)		Y				
Refinery_ProcessingCost	Refinery processing cost	money		Y				
Refinery_ProcessingType	Refinery processing type	varchar(10)		Y				

Table name: DepositWarehouse

Description: The DepositWarehouse table depicts information of warehouse which is mainly used to store the coal product after washing and refinery. It is connected with CoalTargetWarehouse table, by CoalTransportation table which act as a bridge.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
Transportation_Origin_WarehouseID	The original warehouse ID of transportation	int		Y	00001-99999		PK	
Deposit_Warehouse_Capacity	Deposit warehouse capacity	decimal(10, 2)		Y				
Deposit_Warehouse_Temperature	The temperature of deposit warehouse	decimal(5, 2)		Y				
TOW_Manager_ID	Manager ID who is in charge of deposit warehouse	int		Y	0001-9999		FK	ManagerInfor

Table name: RefineryCoal

Description: The table of RefineryCoal mainly depicts the information regarding how to refine and wash the crude coal to add more value onto coal product. According to the regulation and policy of companies, any crude coal must go into refinery factory to purify.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
RefineryCoal_ProductID	The product ID of refinery coal	int		Y	000001-999999		PK	
RefineryCoalProduct_Name	The product name of refinery coal	varchar(40)		Y				
RefineryCoalProduct_Quality	Refinery coal quality	varchar(30)		Y				
RefineCoalProduct_Quantity	Refinery coal quantity	decimal(12, 2)		Y				
RF_CrudeCoal_ProductID	Crude coal product ID	int		Y	00001-99999		FK	CrudeCoal
RF_Refinery_processingID	Refinery coal processing ID	int		Y	0001-9999		FK	RefineryProcessing
RF_Manager_ID	Manager ID	int		Y	0001-9999		FK	ManagerInfor
RF_Transportation_Origin_WarehouseID	Original warehouse ID of transportation	int		Y	00001-99999		FK	DepositWarehouse

Table name: CoalTransportation

Description: It mainly depicts the concrete information of every transportation order which includes the transportation method, beginning time and ending time, and so on.

Name	Desc	Type/ Size	Format	Req'd	Domain	Default/ Derived	Key	Referenced Table
CoalTransportationID	Coal transportation ID	int		Y	0000001-9999999		PK	
CT_Transportation_Origin_WarehouseID	Original warehouse ID of transportation	int		Y	00001-99999		FK	DepositWarehouse
CT_Transportation_Target_WarehouseID	Target warehouse ID of transportation	int		Y	00001-99999		FK	CoalTargetWarehouse
CoalTransportation_Style	The style of transportation	Varchar(20)		Y				
Transportation_Quantity	Transportation quantity	Decimal(14,2)		Y				
Transportation_Begining_Time	Beginning time of transportation	datetime	Mm/dd/yy	Y				
Transportation_Endding_Time	Ending time of transportation	datetime	Mm/dd/yy	Y				
CT_RefineryCoal_ProductID	The product ID of refinery coal	int		Y	000001-999999		FK	RefineryCoal
CT_Manager_ID	Manager ID	int		Y	0001-9999		FK	ManagerInfor

2.2. System catalog

2.2.1. Tables

Object name: CoalBuyerInfo

Object Type: Table

Description: The CoalBuyerInfo table depicts detailed demographic information regarding buyers whose identity are companies.

Detailed Description:

Source: The Forms of sales departments.

Destination: The CoalBuyerInfo table will be the primary source in knowing about the detail of buyers. There will be:

- The CoalBuyerID is the unique identifier which is used in differentiating all coal product buyers.
- All records will be retained on file for ever, because it is beneficial to keep a close relationship with buyers, and all the buyers are valuable resources for companies. Never drop off them.
- Approximately 800 records
- 5% growth per year
- Full access for sales departments. Partial access for human resources departments.
- Related tables include:
    - CoalSaleOrder
    - CoalBuyerCredit

Object name: Coal_FundGathering

Object Type: Table

Description: The Coal_FundGathering table mainly depicts whether the coal payments coming from patrons have been gathered or received.

Detailed Description:

Source: The sales order tracking forms of sales department

Destination: The Coal_FundGathering table will the primary source for all the information of whether the payments coming from patrons are received or not. There will be:

- The CoalSale_OrderID is the unique identifier in differentiating sale orders, and the fund of them is received or not.

- All records will remain on files for 5 years after the fund of sales order has been received.
- Approximately 1600 records
- There is a 500 new records growth per year. Simultaneously there is a nearly 500 old records discarded.
- Full access for sales department and accounting department.
- Related tables include:
    - CoalSaleOrder

Object name: CoalBuyerCredit

Object Type: Table

Description: The CoalBuyerCredit table mainly depicts the credit level of coal buyers. This is made as a warrant or foundation for payments.

Detailed Description:

Source: The credit record form of sales or accounting department

Destination: The CoalBuyerCredit table will be the primary source for the credit data regarding any buyers who are company's existing or potential patrons. There will be:

- The CoalBuyerID is the unique identifier in differentiating tcoal buyers.
- All records will remain on files for 5 years after business relationships are over.
- Approximately 800 records
- There is 5% growth of new buyer's records per year. Simultaneously there is 5% old buyer' records discarded per year.
- Full access for Sales departments
- Related tables include:
    - CoalBuyerInfo

Object name: ManagerInfor

Object Type: table

Description: The ManagerInfor table mainly depicts the concrete information of all managers who have great responsibilities on the supply chain.

Detailed Description:

Source: The forms coming from human recourses departments.

Destination: The ManagerInfor table will be the primary source for all demographic data regarding the managers who are in

charge of crude coal production, refinery coal production, transportation, warehouse, sale order. There will be:

- The Menagerie is the unique identifier which is used to differentiate all managers who undertake different tasks.
- All records will be retained on file for 5 year after employee leaving.
- Approximately 20 records
- There is an unforeseeable increase or decrease on the number of the managers, depending on the economic situation and strategy. However, in normal situation, there is a change under 5% per year.
- Full access for Human recourses department. Partial access for crude coal production, refinery coal production, transportation, warehouse management, and sales departments.
- Related tables include:
    - CrudeCoal
    - RefineryCoal
    - CoalTransportation
    - DepositWarehouse
    - CoalSaleOrder

Object name: CoalTargetWarehouse

Object Type: Table

Description: The table of CoalTargetWarehouse mainly depicts the warehouse information of the buyers. This warehouse usually is the coal transportation target which is used to store the coal product sold to buyers. These kinds of ware house usually belong to users or patrons.

Detailed Description:

Source: The sales, transportation and customer service forms.

Destination: The table of CoalTargetWarehouse mainly contains the information regarding buyers' warehouse. There will be:

- The sales, transportation department will be the main users of the table of CoalTargetWarehouse.
- Transportation_Target_WarehouseID will be the unique identifier which is used to differentiate the different warehouses of the buyers.
- All records will be retained for 5 years.
- Approximately 5000 records.
- There will be approximately a 1000 new records increasing, simultaneously 1000 old ones discarded, per year.
- Full access for transportation and sales departments
- Related tables include:

- CoalSaleOrder
- CoalTransportation

Object name: CoalSaleOrder

Object Type: Table

Description: The CoalSaleOrder table depicts the detailed information of sale orders, highlight on who the buyer is, what the buyers' requirement is, and where buyers want the coal to be transported.

Detailed Description:

Source: The forms of sales department, or the agreement between buyers and companies.

Destination: The CoalSaleOrder table will be the primary source for the information relating with coal sales. There will be:

- The CoalSale_OrderID is the unique identifier in differentiating sales order or agreement on coal product.
- All records will remain on files for 5 years.
- Approximately 3000 records
- There are 300-500 new sales records increasing per year. However, there are also 300-500 old sales records are discarded per year.
- Full access for transportation, warehouse, sales department. Partial access for human resources department.
- Related tables include:
  - CoalTargetWarehouse
  - ManagerInfor
  - CoalBuyerInfo
  - Coal_FundGathering

Object name: CoalMineInfo

Object Type: Table

Description: The CoalMineInfo table contains basic information description regarding coal mines.

Detailed Description:

Source: The form of coal mine information

Destination: The CoalMineInfo mainly describes the information related with coal mine itself. There will be:

- The coal mine management department and coal-producing department will be the main users of this table.
- All of the records will be preserved until the company terminated.
- There are five coal mine's information. With the expansion of the company, new coal mine's information will be added in any time.
- The information of Coal mine usually has no abrupt changes, except the event, such as hiring or firing employees in large scale, and new reserves (coal mine) purchasing.
- Growing with the expansion of companies.
- Full access for coal mine management department and coal-producing department.
- Related tables
    - CrudeCoal

Object name: CrudeCoal

Object Type: Table

Description: The table of CrudeCoal contains the information related with crude coal production.

Detailed Description: **Source:** The crude coal production log, CoalMineInfo table, RefineryCoal table, ManagerInfo table

Destination: The CrudeCoal mainly contains the information regard to crude coal production. There will be:

- The coal production ID will be the unique identifier in differentiating the crude coal production of a fixed period, which is usually one day.
- All records will be retained for 5 years
- Approximately 9000 records
- There will be an 1800 new records increasing, simultaneously 1800 old ones discarded, per year.
- Full access for coal-producing department, coal refinery department. Partial access to coal mine resources management department, and human recourses dept.
- Related tables include:
    - CoalMineInfo
    - CrudeCoal
    - RefineryCoal
    - ManagerInfor

Object name: RefineryProcessing

Object Type: Table

Description: The table of RefineryProcessing mainly depicts the information of refining process which includes processing type, time, and cost per unit.

Detailed Description:

Source: The refinery denotation and illustration forms

Destination: The table of RefineryProcessing contains the information regarding refining process. There will be:

- Coal production and Refinery department will be the main users of the table of RefineryProcessing.
- Refinery_ProcessingID is the unique identifier which is used to differentiate the different kinds of processing on the aspect of coal refining.
- The records of RefineryProcessing table will be retained for ever, until some new refinery processing methods are adopted or some old ones are discarded.
- Approximately 3 records
- In 10 years future, there may be one new records increased, meaning one new coal refining method or technique are invented.
- Full access for refinery coal production department and coal technique development department
- Related table include: RefineryCoal

Object name: DepositWarehouse
Object Type: Table

Description: The DepositWarehouse table depicts information of warehouse which is mainly used to store the coal product after washing and refinery. It is connected with CoalTargetWarehouse table through CoalTransportation table.

Detailed Description:

Source: The forms of the storage management department, sales department, and refinery coal production departments.

Destination: The DepositWarehouse table is the primary resources regarding warehouse used to store the coal product after refinery. There will be:

- Transportation_Origin_WarehouseID will be the unique identifier which is used to differentiate the warehouses used in storing refinery coal.

- All records will be retained on file for a long time until there are new warehouses built, or some new changes made on existing warehouses.
- Approximately about 5 records
- There is basically no change on the records per year, unless new building project relating with warehouse, come into effect.
- Full access for storage management departments, sales departments, and refinery coal departments.
- Related table include:
    - CoalTransportation table
    - RefineryCoal table
    - ManagerInfor table

Object name: RefineryCoal

Object Type: Table

Description: The table of RefineryCoal mainly depicts the information regarding how to refine and wash the crude coal to add more value onto coal product. According to the regulation and policy of companies, any crude coal must go into refinery factory to purify.

Detailed Description: **Source:** The forms or logs of coal refinery factory. CrudeCoal table, RefineryProcessing table, CoalTransportation table, ManagerInfor table, DepositWarehouse table

Destination: The table of RefineryCoal will be the main tools to depict coal refinery information. There will be:

- The RefineryCoal_ProductID will be the unique identifier to differentiate the refinery coal product produced in a fixed period or time interval.
- All records will be retained for 5 years
- Approximately 5000 records
- There will be approximately a 1000 new records increasing, simultaneously 1000 old ones discarded, per year.
- Full access for coal production management department, transportation department, deposit warehouse department. Partial access to HR departments.
- Related tables include:
    - CrudeCoal
    - RefineryProcessing
    - CoalTransportation
    - ManagerInfor
    - DepositWarehouse

Object name: CoalTransportation

Object Type: Table

Description: It mainly depicts the concrete information of every transportation order which includes the transportation method, beginning time and ending time, and so on.

Detailed Description:

Source: The transportation and sales forms.

Destination: The CoalTransportation table is the node connecting different warehouses, such as deposit ware house and transportation target warehouse. There will be:

- CoalTransportationID is the unique identifier which is used to differentiate the different transportation orders.
- All records will be retained for 5 years after its generation
- Approximately 6000 records
- There are 1200 new records added per year, at the same time 1200 order ones are discarded.
- Full access for transportation and sales departments
- Related table include:
    - RefineryCoal
    - CoalTargetWarehouse
    - DepositWarehouse
    - ManagerInfor

2.2.2. Query

Object name: Query_CoalMine_Product_Information

Object Type: Query

Description: This query creates a temporary table for all coal mine information and related crude coal product information.

Detailed Description:

Source: CoalMineInfo, CrudeCoal, Requested Coalmine ID

Destination:

Type: Total query

The Query_CoalMine_Product_Information query creates a table that includes all information from the table of CoalMineInfo and CrudeCoal.

Object name: Query_CrudeCoalProduction_Above60000_Perday

Object Type: Query

Description: This query creates a temporary table for the information of Crude coal which has a production 60000 ton per day, and related coal mine information

Detailed Description:

Source: CoalMineInfo table, CrudeCoal table, Requested coal mine ID

Destination:

Type: Parameter query- request for above a fixed production

The Query_CrudeCoalProduction_Above60000_Perday creates a table which displays the crude coal production above 60000 ton per day. This query is generated on a request for a production above 60000 ton per day.

Object name: Query_CrudeCoalProduction_Quality_Abolve_levle4

Object Type: Query

Description: This query creates a temporary table for crude coal production data which have a quality above level 4.

Detailed Description:

Source: RefineryCoal table, CrudeCoal table, RefineryProcessing table, Requested coal mine ID, Crude coal product ID

Destination:

Type: Parameter query- request for above a fixed quality level

This query creates a table that includes all data coming from coal mine, and related crude coal, which have a quality above level 4. This query is generated on a request based on specific crude coal product and related coal mine.

Object name: Query Coal_Transportation_begining_04012008

Object Type: Query

Description: This query creates a temporary table for coal transportation information beginning from 04/12/2008

Detailed Description:

Source: CoalTransportation table, CoalTargetWarehouse table, Requested transportation target ware house ID, and transportation beginning time

Destination:

Type: Parameter query- request for a fixed date

The query creates a table that includes all data coming from transportation aspects and target warehouse, beginning from 04/12/2008. This query is generated on the basis after a specific date.

Object name: Query_Coal_Transportation_where_LittleRock

Object Type: Query

Description: This query creates a temporary table for coal transportation information, specifying that the target city is LittleRock.

Detailed Description: **Source:** CoalTransportation, CoalTargetWarehouse, Requested transportation target ware house ID, and the city name which target warehouse locates

Destination:

Type: Parameter query- request for a fixed specific place.

This query creates a temporary table which includes all transportation data and related data for a specific transportation target city. This query is generated on a basis for a specific transportation target city.

Object name: Query_Managers_Age_Above30

Object Type: Query

Description: This query creates a temporary table of manager's information whose age is above 30 years old.

Detailed Description: **Source:** ManagerInfor table, Requested manager ID

Destination:

Type: Parameter query- request for a specific age.

This query creates a temporary table which includes all managers' information and related data for a specific age-group managers. This query is generated on a basis for a specific age-group managers.

Object name: Query_SaleOrder_Quantity_Above_600000

Object Type: Query

Description: This query creates a temporary table for sale order information which has a sales quantity above 600000.

Detailed Description: **Source:** CoalSaleOrder, Requested coal sale order ID

Destination:
Type: Parameter query- request for a specific sale quantity

This query creates a temporary table which includes all sales order information and related data for a sale quantity above 600000. This query is generated on a basis for a specific sale quantity.

Object name: Query_Sales_Orders_Receiving_Payments

Object Type: Query

Description: This query creates a temporary table for sale order information, on which company has received the payments coming from buyers or patrons.

Detailed Description:
Source: CoalSaleOrder table, Coal_FundGathering table
Destination:
Type: Parameter query- request for specific payments

This query creates a temporary table which includes all sales order information and related data, on which companies has received the payments from patrons. This query is generated on a basis for a specific group of companies who have paid the money.

Object name: Query_Sale_Orders_Paid_In_Draft_Form

Object Type: Query

Description: This query creates a temporary table for sale order information, focused on which of patrons have paid the money in the form of Draft.

Detailed Description:
Source: CoalSaleOrder table, Coal_FundGathering table, Requested coal sale order Id, and Coal buyer payment method
Destination:
Type: Parameter query- request for specific payments form

This query creates a temporary table which includes all sales order information and related data, focused on which of patrons pay the money in the form of Draft. This query is generated on a basis for a specific payment method.

Object name: Query_buyers_Who_Clearing_payments

Object Type: Query

Description: This query creates a temporary table of sale order information, searching which of patrons have paid the money in the form of Draft.
Detailed
Description: **Source:** Coal_FundGathering table, CoalSaleOrder table, CoalBuyerInfo table
Destination:
Type: Parameter query- request for a specific method of clearing the payments.

This query creates a temporary table which includes all sales order information and dependent data, in searching which of patrons have cleared the payments of sales order. This query is generated on a basis for a specific payment clearing conditions.

Object name: Query_Which_CoalMine_Product_Users_likely_Want

Object Type: Query

Description: This query creates a temporary table in finding which coalmine's product is the one which users most likely want.
Detailed

Description: **Source**: Coal_FundGathering table, CoalSaleOrder table, CoalTargetWarehouse table, CoalTransportation table, RefineryCoal table, CrudeCoal table, CoalMineInfo table, CoalBuyerInfo table. Requested the most popular coal product

Destination:
Type: Parameter query- request for a specific popular coal product.

This query creates a temporary table which includes all coal mine information and dependent data, in finding which coal mine's product are most popular one. This query is generated on a basis for a specific popular degree.

Object name: Query_Managers_Age_symptom_And_Average_Workyear

Object Type: Query

Description: This query creates a temporary output in finding manager's average, maximum, minimum age and average work years.
Detailed

Description: **Source:** ManagerInfor table
Destination:
Type: Parameter query- request for some specific symptom of managers.

This query creates a temporary output which computes the managers' symptom. This query is generated on a request basis for some specific symptoms of managers.

Object name: Query_Average_Transportation_time

Object Type: Query

Description: This query creates a temporary output in finding average transportation time from original warehouse to target warehouse.

Detailed Description: **Source:** CoalTransportation
Destination:
Type: Parameter query- request for average transportation time.

This query creates a temporary output which computes average transportation time. This query is generated on a request basis for computing average transportation time.

Object name: Query_Managers_whose_Workyear_Above_Average_Level

Object Type: Query

Description: This query creates a temporary table of managers' information regarding managers' work year is above average level.

Detailed Description: **Source:** ManagerInfor table, Requested Manager ID
Destination:
Type: Parameter query- request for a specific qualification of managers

This query creates a temporary table which includes managers whose work year is above average work year level. This query is generated on a basis for a specific work year requirement of managers.

2.2.3. Forms

Object name: Form_CoalBuyerInfo

Object Type: Form

Description: The CoalBuyerInfo form maintains the detailed information relating with coal buyers.

Detailed Description:
Source: CoalBuyerInfo table
Destination: CoalBuyerInfo table

The coal buyer information form is used to enter buyer information into database. Additionally, this form is used for ongoing maintenance, such as coal buyer address, name, etc. The data from coal buyer information form updates the data stored in CoalBuyerInfo table.

Volume: 1 form per buyer
Frequency: not frequently (It generally updates when new buyer build relations with us, or old buyers change their address or contact methods.

Object name: Form_Coal_FundGathering

Object Type: Form

Description: The Coal_FundGathering form maintains the data relating with coal payments coming from patrons.

Detailed Description:
Source: Coal_FundGathering table
Destination: Coal_FundGathering table

The coal fund gathering form is used to enter payments data into the database. Additionally, this form can be used as maintenance, such as payment method, payment final date, and payments status. The data from coal fund gathering form updates the data stored in Coal_FundGathering table.

Volume: 1 from per sales order
Frequency: It is usually happened when the payments from patrons are received, according AICPA standards.

Object name: Form_CoalBuyerCredit

Object Type: Form

Description: The CoalBuyerCredit form maintains the data relating with credits of patrons and buyers.

Detailed Description: **Source:** CoalBuyerCredit table
Destination: CoalBuyerCredit table

The coal buyer credit form is used to enter credit data into database. Additionally, this form is used for ongoing maintenance, such as coal buyer credit level. The data from coal buyer credit form updates the data stored in CoalBuyerCredit table.

Volume: 1 from per coal buyer
Frequency: It is frequent. (Especially at the time that a new patron makes a business with us or old patrons' financial status has a severe change.)

Object name: Form_ManagerInfor

Object Type: Form

Description: The ManagerInfor form maintains the demographic information of managers.

Detailed Description: **Source:** ManagerInfor
Destination: ManagerInfor

The Manager Information form is used to enter managers' demographic information into database. Additionally, this form is used for ongoing maintenance, such as manager's job type, department, SSN, etc. The data from Manager Information form updates the data stored in ManagerInfor table.

Volume: 1 form per manager
Frequency: On request (generally happened once per year and per manager)

Object name: Form_CoalTargetWarehouse

Object Type: Form

Description: The CoalTargetWarehouse form maintains the detailed information of target warehouse which usually belong to patrons.

Detailed Description:

Source: CoalTargetWarehouse table
Destination: CoalTargetWarehouse table

The target coal warehouse form is used to enter patrons' warehouse information into database. Additionally, this form is used for ongoing maintenance, such as patrons' coal warehouse address, etc. The data from target coal warehouse form updates the data stored in CoalTargetWarehouse table.

Volume: 1 form per patron's warehouse
Frequency: happened with every patron's transportation request in sales orders.

Object name: Form_CoalSaleOrder

Object Type: Form

Description: The CoalSaleOrder form maintains the detailed information of sales order.

Detailed Description:

Source: CoalSaleOrder
Destination: CoalSaleOrder

The coal sale order form is used to enter sales order information into database. Additionally, this form is used for ongoing maintenance, such as coal sale quantity, coal sale quality requests, delivering time request, etc. The data from coal sale order form updates the data stored in CoalSaleOrder table.

Volume: 1 form per sale order
Frequency: on request (generally happened when new sale order reaches or old orders are in modification.)

Object name: Form_CoalMineInfo

Object Type: Form

Description: The CoalMineInfo form maintains the detailed information of coal mine.

Detailed Description:

Source: CoalMineInfo table

Destination: CoalMineInfo table

The coal mine information form is used to enter coal mine information into database. Additionally, this form is used for ongoing maintenance, such as coal mine address, reserves, how many employees they hire, etc. The data from coal mine information form updates the data stored in CoalMineInfo table.

Volume: 1 form per coal mine
Frequency: on request

Object name: Form_CrudeCoal

Object Type: Form

Description: The CrudeCoal form maintains the detailed information of crude coal production.

Detailed Description: **Source:** CrudeCoal table
Destination: CrudeCoal table

The Crude Coal form is used to enter detailed coal production information into database. Additionally, this form is used for ongoing maintenance, such as crude coal product name, quality, quantity, etc. The data from Crude Coal form updates the data stored in CrudeCoal table.

Volume: 1 for per crude coal product ID (per batch)
Frequency: happened frequently with the coal production process

---

Object name: Form_RefineryProcessing

Object Type: Form

Description: The RefineryProcessing form maintains the data relating with refinery coal processing details.

Detailed Description: **Source:** RefineryProcessing table
Destination: RefineryProcessing table

The Refinery Processing form is used to enter detailed information of refinery processing into database. Additionally, this form is used for ongoing maintenance, such as refinery processing name, time, cost, etc.

The data from Refinery Processing form updates the data stored in RefineryProcessing table.

Volume: 1 form per refinery processing ID
Frequency: frequently happened with the refinery coal production process

---

Object name: Form_DepositWarehouse

Object Type: Form

Description: The DepositWarehouse form maintains the detailed information of company's deposit warehouse.

Detailed Description: **Source:** DepositWarehouse
Destination: DepositWarehouse

The deposit warehouse form is used to enter detailed refinery coal information into database. Additionally, this form is used for ongoing maintenance, such as coal warehouse capacity, temperature, etc. The data from deposit warehouse form updates the data stored in DepositWarehouse table.

Volume: 1 form per origin warehouse (In usual, origin ware house are in the vicinity of coal mine)
Frequency: frequently happened with the refinery coal producing.

Object name: Form_RefineryCoal

Object Type: Form

Description: The RefineryCoal form maintains the data relating with refinery coal processing.

Detailed Description: **Source:** RefineryCoal table
Destination: RefineryCoal table

The refinery coal form is used to enter refinery coal processing information into database. Additionally, this form is used for ongoing maintenance, such as refinery coal product name, quantity, quality, etc. The data from refinery coal form updates the data stored in RefineryCoal table.

Volume: 1 form per refinery coal product ID (per batch)

Frequency: frequently happened with the refinery coal producing.

Object name: Form_CoalTransportation

Object Type: Form

Description: The CoalTransportation form maintains the detailed information of coal transportation.

Detailed Description: **Source:** CoalTransportation table
Destination: CoalTransportation table

The Coal Transportation form is used to enter detailed transportation information into database. Additionally, this form is used for ongoing maintenance, such as original warehouse ID, target warehouse ID, transportation style, transportation time, etc. The data from Coal Transportation form updates the data stored in CoalTransportation table.

Volume: 1 form per transportation ID
Frequency: happened when every patron's transportation request was transferred to transportation department.

2.2.4. Reports

Object name: Rpt_CoalBuyerInfo

Object Type: Report

Description: The Coal buyer information report is a report for the details of coal buyer information

Detailed Description: **Source:** CoalBuyerInfo table
Destination: coal buyer information

Sales department, transportation, and accounting department request coal buyer information reports on an as needed basis. These reports are made not frequently. These reports are updated according to management requirements.

Object name: Rpt_Coal_FundGathering

Object Type: Report

Description: The Coal fund gathering report is a report for the detail of payments.

Detailed Description:
Source: Coal_FundGathering table
Destination: Coal fund gathering information

Sales department and accounting department request coal fund gathering reports on an as needed basis. On every sales order time limit, these reports are made and verified as the basis of decision-making. These reports are updated according to management requirements.

Object name: Rpt_CoalBuyerCredit

Object Type: Report

Description: The Coal buyer credit report is a report for the detail of coal buyer credit information.

Detailed Description:
Source: CoalBuyerCredit table
Destination: coal buyer credit information reports

Sales department and accounting department request coal buyer credit information reports on an as needed basis. In the advance of signing every sales order, these reports are made and verified as the basis of decision-making. These reports are updated according to management requirements.

Object name: Rpt_ManagerInfor

Object Type: Report

Description: The manager information report is a report for the details of demographic information of managers in any post.

Detailed Description:
Source: ManagerInfor table
Destination: Manager demographic information report

Human resources department requires manager’s demographic information on an as needed basis. These reports are made not frequently.

These reports are updated with the managers hiring and firing events happened.

Object name: Rpt_CoalTargetWarehouse

Object Type: Report

Description: The Coal target warehouse report is a report for the details of target warehouse in patron's or buyer's side.

Detailed Description: **Source:** CoalTargetWarehouse
Destination: Coal target ware house report

Transportation department directly requires coal target warehouse information on an as needed basis. These reports are made frequently with every sale order signed and the target warehouse is clarified. These reports are updated with the new sale order signed and old ones adjusting.

Object name: Rpt_CoalSaleOrder

Object Type: Report

Description: The Coal sale order report is a report for the details of coal sale orders

Detailed Description: **Source:** CoalSaleOrder table
Destination: Coal sale order report

Sales department and transportation department directly requires coal sale order information on an as needed basis. These reports are made frequently with the every sale order signed. These reports are updated with the new sale order signing and old ones adjusting.

Object name: Rpt_CoalMineInfo

Object Type: Report

Description: The Coal mine information report is a report for the details of coal mine information

Detailed Description: **Source:** CoalMineInfo table
Destination: coal mine information report

Resources management department requires coal mine information on an as needed basis. These reports are made not frequently. These reports are updated with the new coal mine building and old coal mine adjusting.

Object name: Rpt_CrudeCoal

Object Type: Report

Description: The crude coal report is a report for the details of crude coal production information

Detailed Description: **Source:** CrudeCoal table
Destination: Crude coal production report

Coal production management department requires crude coal information on an as needed basis. These reports are made frequently with the normal producing process in coal mine. These reports are updated with the produced coal going into refinery. (In usual, crude coal immediately enter into refinery factory when produced)

Object name: Rpt_RefineryProcessing

Object Type: Report

Description: The Refinery processing information report is a report for the details of refinery processing for every batch of coal product.

Detailed Description: **Source:** RefineryProcessing
Destination: Refinery processing report

Coal production management department and accounting department require Refining processing information on an as needed basis. These reports are made frequently with the production process in refinery factory.

Object name: Rpt_DepositWarehouse

Object Type: Report

Description: The deposit warehouse report is a report for the details of warehouse information which is near to the coal mine. (In usual, the deposit warehouse are in the vicinity of coal mine and refinery factory.)

Detailed Description: **Source:** DepositWarehouse
Destination: Deposit warehouse report

Coal production management department and warehouse management department requires deposit warehouse information on an as needed basis. These reports are made frequently with the normal coal producing process in refinery factory. These reports are updated with the refined coal going from refinery into deposit warehouse.

Object name: Rpt_RefineryCoal

Object Type: Report

Description: The Refinery Coal information report is a report for the details of refinery coal in refinery factory.

Detailed Description: **Source:** RefineryCoal table
Destination: Refining coal report

Coal production management department requires Refining coal information on an as needed basis. These reports are made frequently with the production process in refinery factory. These reports are updated with refinery coal produced from refinery factory.

Object name: Rpt_CoalTransportation

Object Type: Report

Description: The Coal transportation report is a report for the details of how to transport coal product from original warehouse to patron’s warehouse.

Detailed Description: **Source:** CoalTransportation table
Destination: coal transportation report

Transportation department directly requires coal target warehouse information on an as needed basis. These reports are made frequently with

the every sale order signed and the target warehouse is clarified. These reports are updated with the new sale order signed and old ones adjusting.

---

3. Lessons learned

3.1. Five lessons

The first, during ERD design phrase, it is important to build a bridge entity between many to many relationships. The second, it is important to use the code to represent some field value for the convenience of query in future. The third, it is very important to make a deep analysis on the topic or project before building database. The fourth, it is very important to meditate and analyze what kinds of business rules database model should support. The fifth, it is important to extract entities or attributes in 3 rd normal form.

3.2. The relationship between project and class materials

- The analysis in project employs the business rules analysis methods mentioned in class.
- During the process of building Entity Relationship Diagrams, the method of creating ERD mentioned in class is employed.
- The method of creating complex ERD mentioned in class is employed.
- During the process of building database, the method of normalization is employed.
- During the process of building database and query, the SQL commands and syntax (Lans, 2007) mentioned in class are employed, such as creating table, inserting records, display records, deleting records, dropping tables, nest queries, scalar functions, column functions, and grouping data, etc.
- The concepts of database backup and recovery mentioned in class are employed.
- The concepts of the responsibilities of database administrator mentioned in class are employed.
- Distributed DBMS system involving the combination of SQL 2003 and ACCESS mentioned in class is employed. The input forms and output reports are generated by using ACCESS for its convenient use.
- Many other concepts, methods, and thinking mentioned in class are the foundation and basis in completing this project. They are too much more to list one by one.

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5. Appendices

5.1. The SQL code of creating tables

```
--Create Table Collection Total
--Pay attention to the order of building table
--1.CoalBuyerInfo
--2.Coal_FundGathering
--3.CoalBuyerCredit
--4.ManagerInfor
--5.CoalTargetwarehouse
--6.CoalSaleOrder
--7.CoalMineInfo
--8.CrudeCoal
--9.RefineryProcessing
--10.DepositWarehouse
--11.RefineryCoal
--12.CoalTransportation


--1.create table CoalBuyerInfo
create table CoalBuyerInfo
(
      CoalBuyerID                       int,
      CoalBuyer_Name                    Varchar(30),
      CoalBuyer_OfficeNumber        Varchar(10),
      CoalBuyer_StreetName          Varchar(30),
      CoalBuyer_CityName                Varchar(30),
      CoalBuyer_StateName               Varchar(30),
      CoalBuyer_CountryName         Varchar(30),
      CoalBuyer_Zip                     varchar(10),
      Constraint  CoalBuyerID_PK    primary key (CoalBuyerID)
);

--The end of create table CoalBuyerInfo
```

```
--2.Create table Coal_FundGathering
Create table Coal_FundGathering
(
      CF_CoalSale_OrderID                        int,
      CoalBuyer_PaymentsMethod             varchar(10) check
(CoalBuyer_PaymentsMethod in ('1','2','')),
      CoalBuyer_PaymentsFinalDate          datetime  ,
      CoalBuyer_PaymentsStatus             Varchar(10) check
(CoalBuyer_PaymentsStatus in ('1','0')),
      Constraint  CF_CoalSale_OrderID_PK  primary key
      (CF_CoalSale_OrderID),
);

--The end of Create table Coal_FundGathering


--3.Create table CoalBuyerCredit

Create table CoalBuyerCredit
(
      CBC_CoalBuyerID               int,
      CoalBuyer_Credit_Level  varchar(10),
      Constraint  CBC_CoalBuyerID_PK     primary key (CBC_CoalBuyerID),
      Constraint  CBC_CoalBuyerID_FK      foreign key (CBC_CoalBuyerID)
            references  CoalBuyerInfo(CoalBuyerID),
);

--The end of create table CoalBuyerCredit


--4.Create the table of ManagerInfor
Create table ManagerInfor
(
      Manager_ID              int,
      Manager_JobType         varchar(40),
      Manager_Department      varchar(40),
      Manager_FirstName varchar(30),
      Manager_MiddleName      varchar(20),
      Manager_LastName  varchar(30),
      Manager_Sex             varchar(1),
      Manager_Birthday  datetime,
      Manager_WorkYears int,
      Manager_SSN             varchar(9),
      Constraint  Manager_ID_PK     primary key (Manager_ID),
      Constraint  Manager_Sex_CK    check (Manager_Sex=0 or
Manager_Sex=1),
      Constraint  Manager_WorkYears_CK    check (Manager_WorkYears>0
and Manager_WorkYears<=99),
      Constraint  Manager_SSN_CK                Check
      (len(Manager_SSN)=9),
);

--The end of Create the table of ManagerInfor


--5.create table CoalTargetWarehouse
```

```
Create table CoalTargetWarehouse
(
      Transportation_Target_WarehouseID                    int,
      Transportation_Target_WarehouseName                  Varchar(30),
      Transportation_Target_Warehouse_StreetName           varchar(50),
      Transportation_Target_Warehouse_CityName       Varchar(30),
      Transportation_Target_Warehouse_StateName      Varchar(10),
      Transportation_Target_Warehouse_CountryName          Varchar(30),
      Transportation_Target_Warehouse_Zip                  Varchar(10),
      Trans_CoalSale_OrderID                                    int,
      Constraint Transportation_Target_WarehouseID_PK primary key
(Transportation_Target_WarehouseID),
);

--The end of Create table CoalTargetWarehouse


--6. Create table CoalSale_OrderID
Create table CoalSaleOrder
(
      CoalSale_OrderID       int,
      CS_CoalBuyerID               int,
      CoalSale_Quantity      decimal(14,2),
      CoalSale_QualityRequest varchar(10),
      CoalSale_Deliver_Time  datetime,
      CS_Manager_ID                int,
      CS_Transportation_Target_WareHouseID     int,
      Constraint  CoalSale_OrderID_PK     primary key
(CoalSale_OrderID),
      Constraint  CS_CoalBuyerID_FK foreign key (CS_CoalBuyerID)
            references  CoalBuyerInfo(CoalBuyerID),
      Constraint  CS_Transportation_Target_WareHouseID_FK   foreign key
(CS_Transportation_Target_WareHouseID)
            references
      CoalTargetWarehouse(Transportation_Target_WarehouseID),
      Constraint        CS_Manager_ID_FK  foreign key (CS_Manager_ID)
            references ManagerInfor(Manager_ID)
);

--The end of Create table CoalSale_OrderID


--7. Create Table CoalMineInfo
--It demonstrate the concrete information of Coal mine, such as
location, state, country
--, reserve quantities, and the number of employees hired by coalmine.

Create table CoalMineInfo
(
      CoalMine_ID                         int,
      CoalMine_Place                      varchar(60),
      CoalMine_State                      varchar(20),
      CoalMine_Country              varchar(30),
      CoalMine_Reserves             decimal(15,2),
      CoalMine_EmployeeNumber       int,
      Constraint CoalMine_ID_PK     primary key (CoalMine_ID),
      Constraint CoalMine_Reserves_CK     Check (CoalMine_Reserves>=0),
```

```
      Constraint CoalMine_EmployeeNumber_CK Check
(CoalMine_EmployeeNumber>=0)
);

--The end of Creat Table CoalMineInfo


--8. Create Table CrudeCoal
--It depicts the concrete information about the crude coal, such as the
time of digging,
--the name of crude coal, the quality level of crude coal, the
production of coal, and the manager responsible for production.

Create table CrudeCoal
(
      CrudeCoal_ProductID                int,
      CrudeCoalProducingTime        datetime,
      CrudeCoalProduct_Name         varchar(50),
      CrudeCoalProduct_Quality      varchar(30),
      CrudeCoalProduct_Quantity     decimal(12,2),
      CrudeCoal_CoalMineID          int,
      CrudeCoal_ManagerID                int,

      Constraint CrudeCoal_ProductID_PK primary key
(CrudeCoal_ProductID),
      Constraint CrudeCoal_CoalMineID_FK foreign key
(CrudeCoal_CoalMineID)
            references CoalMineInfo(CoalMine_ID),
      Constraint CrudeCoal_ManagerID_FK foreign key
(CrudeCoal_ManagerID)
            references ManagerInfor(Manager_ID)
);

--The end of Create Table CrudeCoal


--9. Create the table of RefineryProcessing
Create table RefineryProcessing
(
      Refinery_ProcessingID   int,
      Refinery_ProcessingName Varchar(50),
      Refinery_ProcessingCost money,
      Refinery_ProcessingType varChar(10),
      Constraint  Refinery_ProcessingID_PK      primary key
(Refinery_ProcessingID),
);

--The end of Create the table of RefineryProcessing


--10. Create the table of DepositWarehouse
create table DepositWarehouse
(
      Transportation_Origin_WarehouseID   int,
      Deposit_Warehouse_Capacity                decimal(10,2),
      Deposit_Warehouse_Temperature       decimal(5,2),
      TOW_Manager_ID                                int,
```

```
      Constraint  Transportation_Origin_WarehouseID_PK      primary key
      (Transportation_Origin_WarehouseID),
      Constraint  TOW_Manager_ID_FK foreign key(TOW_Manager_ID)
            references  ManagerInfor(Manager_ID),
);
--The end of Create the table of DepositWarehouse


--11. create the table RefineryCoal
Create table RefineryCoal
(
      RefineryCoal_ProductID              int,
      RefineryCoalProduct_Name                  varchar(40),
      RefineryCoalProduct_Quality         varchar(30),
      RefineCoalProduct_Quantity          decimal(12,2),
      RF_CrudeCoal_ProductID              int,
      RF_Refinery_processingID            int,
      RF_Manager_ID                             int,
      RF_Transportation_Origin_WarehouseID      int,

      Constraint  RefineryCoal_ProductID_PK     primary key
(RefineryCoal_ProductID),
      Constraint  RF_CrudeCoal_ProductID_FK     foreign key
(RF_CrudeCoal_ProductID)
                  references CrudeCoal(CrudeCoal_ProductID),
      Constraint  RF_Refinery_processingID_FK   foreign key
      (RF_Refinery_processingID)
                  references RefineryProcessing(Refinery_ProcessingID),
      Constraint  RF_Manager_ID_FK                    foreign key
      (RF_Manager_ID)
                  references ManagerInfor(Manager_ID),
      Constraint  RF_Transportation_Origin_WarehouseID_FK foreign key
      (RF_Transportation_Origin_WarehouseID)
                  references
      DepositWarehouse(Transportation_Origin_WarehouseID),
);
--The end of Creating the table RefineCoal


--12. Create the table CoalTransportation
Create table CoalTransportation
(
      CoalTransportationID                      int,
      CT_Transportation_Origin_WarehouseID      int,
      CT_Transportation_Target_WarehouseID      int,
      CoalTransportation_Style                  varchar(20),
      Transportation_Quantity                   decimal(14,2),
      Transportation_Begining_Time        datetime,
      Transportation_Endding_Time               datetime,
      CT_RefineryCoal_ProductID                       int,
      CT_Manager_ID                                         int,
      Constraint  CoalTransportationID_PK primary key
      (CoalTransportationID),
      Constraint  CT_Transportation_Origin_WarehouseID_FK foreign key
(CT_Transportation_Origin_WarehouseID)
            references
      DepositWarehouse(Transportation_Origin_WarehouseID),
```

```
      Constraint  CT_Transportation_Target_WarehouseID_FK foreign key
(CT_Transportation_Target_WarehouseID)
            references
      CoalTargetWarehouse(Transportation_Target_WarehouseID),
      Constraint  CT_RefineryCoal_ProductID_FK  foreign key
      (CT_RefineryCoal_ProductID)
            references  RefineryCoal(RefineryCoal_ProductID),
      Constraint  CT_Manager_ID_FK  foreign key (CT_Manager_ID)
            references  ManagerInfor(Manager_ID)
);
--The end of create the table CoalTransportation

----The end of Create Table Collection Total
```

5.2. **The Query code and results**

```
--1.Quer_CoalMine_Product_Information
--2.Query_CrudeCoalProduction_Above60000_Perday
--3.Query_CrudeCoalProduction_Quality_Abolve_levle4
--4. Query_Coal_Transportation_begining_04012008
--5. Query_Coal_Transportation_where_LittleRock
--6.Query_Managers_Age_Above3o
--7.Query_SaleOrder_Quantity_Above_600000
--8.Query_Sales_Orders_Receiving_Payments
--9.Query_Sale_Orders_Paid_In_Draft_Form
--10.Query_buyers_Who_Clearing_payments
--11.Query_Which_CoalMine_Product_Users_likely_Want
--12. Query_Managers_Age_symptom_And_Average_Workyear
--13. Query_Average_Transportation_time
--14. Query_Managers_whose_Workyear_Above_Average_Level


--1.Quer_CoalMine_Product_Information
--Query the coal mine information and the production information
simutaneously

Select
CrudeCoal_ProductID,CrudeCoalProduct_Name,CoalMine_ID,CoalMine_Place
      from CoalMineInfo, CrudeCoal
            where CrudeCoal_CoalMineID=CoalMine_ID
--The end of Query the coal mine information and the production
information simutaneously


--2.Query_CrudeCoalProduction_Above60000_Perday
--Query the crude coal production which is above 60000 ton one day
--Simutaneously display the coalmine

Select
CrudeCoal_ProductID,CrudeCoalProduct_Name,CrudeCoalProduct_Quality,
CrudeCoalProduct_Quantity, CoalMine_ID,CoalMine_Place
                  from CoalMineInfo, CrudeCoal
                        where CrudeCoal_CoalMineID=CoalMine_ID and
CrudeCoalProduct_Quantity>=70000
--The end of Query the crude coal production which is above 60000 ton
one day
```

```
--3.Query_CrudeCoalProduction_Quality_Abolve_levle4
--Query crude coal production status,which quality is equal or above
level 4, and raw coal they used,
--and the related processing information.

Select RefineryCoal_ProductID,
RefineryCoalProduct_Name,RefineryCoalProduct_Quality,RefineCoalProduct_
Quantity,CrudeCoal_ProductID,CrudeCoalProduct_Name,Refinery_ProcessingI
D,Refinery_ProcessingName
      from RefineryCoal,CrudeCoal,RefineryProcessing
            where RF_Refinery_processingID=Refinery_ProcessingID and
RF_CrudeCoal_ProductID=CrudeCoal_ProductID and
RefineryCoalProduct_Quality>=4
--The end of Query crude coal production status,which quanlity is equal
or above level 4, and raw coal they used,


--4. Query_Coal_Transportation_begining_04012008
--Query coal transportation information which is beginning after the
date of 04/01/2008

Select
CoalTransportationID,CT_Transportation_Origin_WarehouseID,CT_Transporta
tion_Target_WarehouseID,Transportation_Target_Warehouse_StreetName,Tran
sportation_Target_Warehouse_CityName,Transportation_Target_Warehouse_St
ateName,Transportation_Target_Warehouse_CountryName,Transportation_Targ
et_Warehouse_CountryName,Transportation_Target_Warehouse_Zip,CoalTransp
ortation_Style,Transportation_Quantity,Transportation_Begining_Time,Tra
nsportation_Endding_Time,CT_RefineryCoal_ProductID
      from CoalTransportation, CoalTargetWarehouse
            where
CT_Transportation_Target_WarehouseID=Transportation_Target_WarehouseID
and Transportation_Begining_Time>='04/01/2008'
--The end of Query coal Transportation information


--5. Query_Coal_Transportation_where_LittleRock
--Query coal transportation information which of target is the city
LittleRock

Select
CoalTransportationID,CT_Transportation_Origin_WarehouseID,CT_Transporta
tion_Target_WarehouseID,Transportation_Target_Warehouse_StreetName,Tran
sportation_Target_Warehouse_CityName,Transportation_Target_Warehouse_St
ateName,Transportation_Target_Warehouse_CountryName,Transportation_Targ
et_Warehouse_CountryName,Transportation_Target_Warehouse_Zip,CoalTransp
ortation_Style,Transportation_Quantity,Transportation_Begining_Time,Tra
nsportation_Endding_Time,CT_RefineryCoal_ProductID
      from CoalTransportation, CoalTargetWarehouse
            where
CT_Transportation_Target_WarehouseID=Transportation_Target_WarehouseID
and Transportation_Target_Warehouse_CityName='LittleRock'
--The end of Query coal transportation information which of target is
the city little rock
```

```
--6.Query_Managers_Age_Above3o
--Query the managers whose age is above 30

Select
Manager_ID,Manager_JobType,Manager_Department,Manager_FirstName,Manager
_MiddleName,Manager_LastName,Manager_Sex,Manager_Birthday,Manager_WorkY
ears,Manager_SSN
      from ManagerInfor
            where (year(getdate())-year(Manager_Birthday))>30
--The end of Query the managers whose age is above 30


--7.Query_SaleOrder_Quantity_Above_600000
--Query the information of CoalSaleOrder in which sale quantity is
equal and above 6000000.

select
CoalSale_OrderID,CS_CoalBuyerID,CoalSale_Quantity,CoalSale_QualityReque
st,CoalSale_Deliver_Time,CS_Manager_ID,CS_Transportation_Target_WareHou
seID
      from CoalSaleOrder
            where CoalSale_Quantity>=6000000;
--The end of Query the information of CoalSaleOrder in which sale
quantity is equal and above 6000000.


--8.Query_Sales_Orders_Receiving_Payments
--Query the sale orders which have received the payments from customers
or patrons.

select
CoalSale_OrderID,CS_CoalBuyerID,CoalSale_Quantity,CoalSale_QualityReque
st,CoalSale_Deliver_Time,CS_Manager_ID,CS_Transportation_Target_WareHou
seID
      from CoalSaleOrder, Coal_FundGathering
            where CF_CoalSale_OrderID=CoalSale_OrderID and
CoalBuyer_PaymentsStatus=1;
--The end of Query the sale orders which have received the payments
from customers or patrons.


--9.Query_Sale_Orders_Paid_In_Draft_Form
--Query the sale orders which patron prepares to pay in draft form.

select
CoalSale_OrderID,CS_CoalBuyerID,CoalSale_Quantity,CoalSale_QualityReque
st,CoalSale_Deliver_Time,CS_Manager_ID,CS_Transportation_Target_WareHou
seID,CoalBuyer_PaymentsMethod
      from CoalSaleOrder, Coal_FundGathering
            where CF_CoalSale_OrderID=CoalSale_OrderID and
CoalBuyer_PaymentsMethod=2;
--The end of Query the sale orders which patron prepares to pay in
draft form.


--10.Query_buyers_Who_Clearing_payments
----Query the buyer information who have cleared the payments
```

```
select
CoalBuyerID,CoalBuyer_Name,CoalBuyer_OfficeNumber,CoalBuyer_StreetName,
CoalBuyer_CityName,CoalBuyer_StateName,CoalBuyer_CountryName,CoalBuyer_
Zip
      from Coal_FundGathering, CoalSaleOrder, CoalBuyerInfo
            where CF_CoalSale_OrderID=CoalSale_OrderID and
CS_CoalBuyerID=CoalBuyerID and CoalBuyer_PaymentsStatus=1
--The end of Query the buyer information who have clear the payments


--11.Query_Which_CoalMine_Product_Users_likely_Want
--Query which coalmine produces the coal which buyers would like
quickly to clear the payments

select
CoalMine_ID,CoalMine_Place,CoalMine_State,CoalMine_Country,CoalMine_Res
erves,CoalMine_EmployeeNumber,CoalBuyer_Name
      from Coal_FundGathering,
CoalSaleOrder,CoalTargetWarehouse,CoalTransportation,RefineryCoal,Crude
Coal, CoalMineInfo, CoalBuyerInfo
            where CF_CoalSale_OrderID=CoalSale_OrderID and
CS_Transportation_Target_WareHouseID=Transportation_Target_WarehouseID
and
Transportation_Target_WarehouseID=CT_Transportation_Target_WarehouseID
and CT_RefineryCoal_ProductID=RefineryCoal_ProductID and
RF_CrudeCoal_ProductID=CrudeCoal_ProductID and
CrudeCoal_CoalMineID=CoalMine_ID and CS_CoalBuyerID=CoalBuyerID and
CoalBuyer_PaymentsStatus=1
--The end of Query which coalmine produces the coal which buyers would
like quickly to clear the payments


--12. Query_Managers_Age_symptom_And_Average_Workyear
--Compute the average age, max age, min age, average workyear,
--max workyear, and min workyear of managers
--and display its detailed information

select avg(year(getdate())-year(Manager_Birthday)) 'Average age',
max(year(getdate())-year(Manager_Birthday))'Oldest
age',min(year(getdate())-year(Manager_Birthday)) 'Smallest age',
avg(Manager_WorkYears) 'Average work
years',Max(Manager_WorkYears)'Longest work years',min(Manager_WorkYears)
'Shortest work years'
      from ManagerInfor
--The end of --Compute the average age, max age, min age, average
workyear,

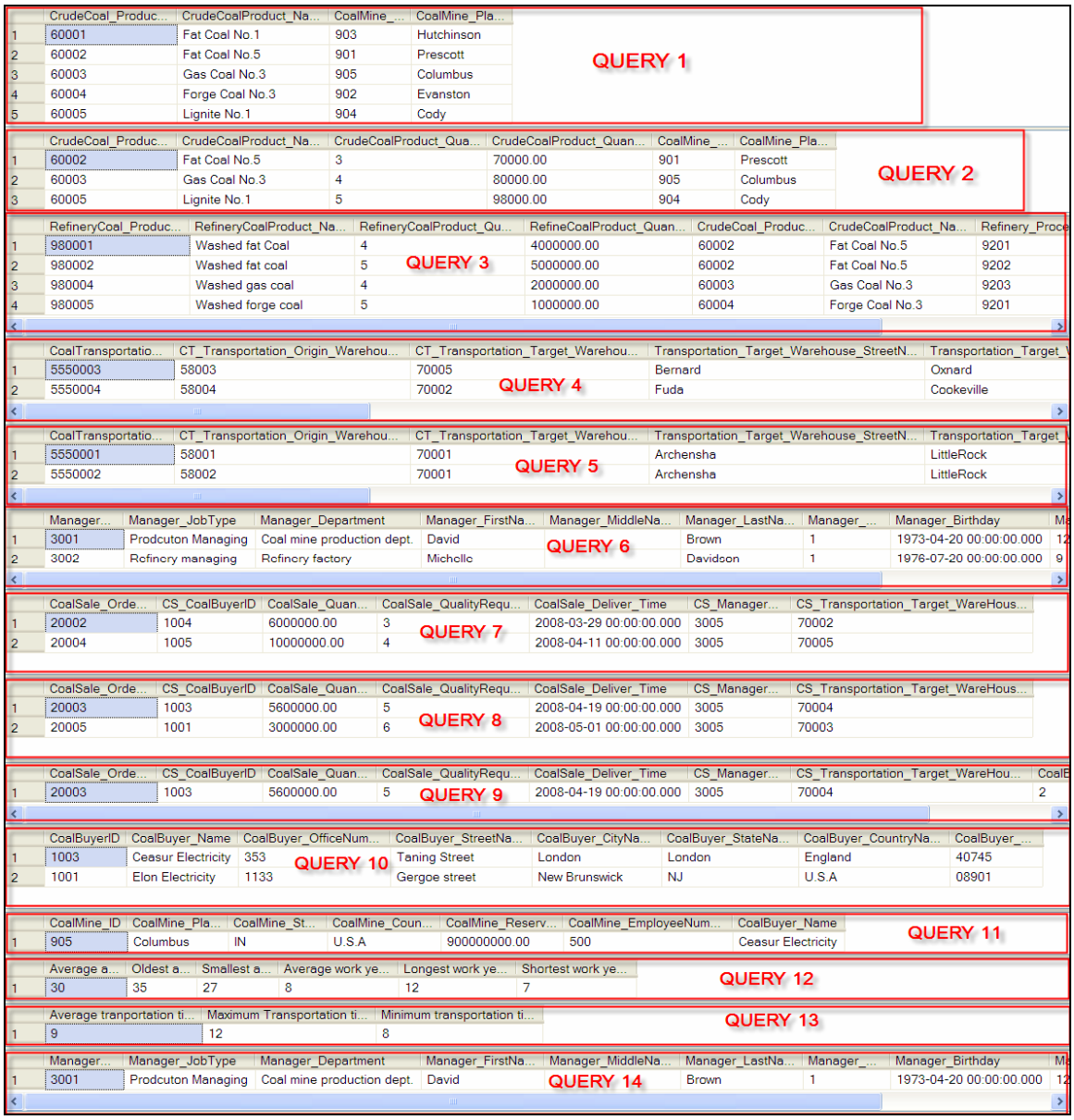

--13. Query_Average_Transportation_time
--Compute the Average transportation time from original warehouse to
target ones,
-- Max transportation time, and min transportation time

Select
avg(datediff(day,Transportation_Begining_Time,Transportation_Endding_Ti
me )) 'Average tranportation time',
```

```
max(datediff(day,Transportation_Begining_Time,Transportation_Endding_Ti
me )) 'Maximum Transportation time',
min(datediff(day,Transportation_Begining_Time,Transportation_Endding_Ti
me)) 'Minimum transportation time'
      from CoalTransportation
--The end of Compute the Average transportation time from original
warehouse to target ones,


--14. Query_Managers_whose_Workyear_Above_Average_Level
--Pick up whose work-year is above the average level,
--and display its detailed information

select top 20 percent
Manager_ID,Manager_JobType,Manager_Department,Manager_FirstName,Manager
_MiddleName,Manager_LastName,Manager_Sex,Manager_Birthday,Manager_WorkY
ears,Manager_SSN
      from ManagerInfor
            where Manager_WorkYears>(select AvG(Manager_WorkYears) from
ManagerInfor)
--The end of Pick up whose work-year is above the average level.
```



QUERY 1

	CrudeCoal_Produc...	CrudeCoalProduct_Na...	CoalMine_...	CoalMine_Pla...
1	60001	Fat Coal No.1	903	Hutchinson
2	60002	Fat Coal No.5	901	Prescott
3	60003	Gas Coal No.3	905	Columbus
4	60004	Forge Coal No.3	902	Evanston
5	60005	Lignite No.1	904	Cody

QUERY 2

	CrudeCoal_Produc...	CrudeCoalProduct_Na...	CrudeCoalProduct_Qua...	CrudeCoalProduct_Quan...	CoalMine_...	CoalMine_Pla...
1	60002	Fat Coal No.5	3	70000.00	901	Prescott
2	60003	Gas Coal No.3	4	80000.00	905	Columbus
3	60005	Lignite No.1	5	98000.00	904	Cody

QUERY 3

	RefineryCoal_Produc...	RefineryCoalProduct_Na...	RefineryCoalProduct_Qu...	RefineCoalProduct_Quan...	CrudeCoal_Produc...	CrudeCoalProduct_Na...	Refinery_Proce
1	980001	Washed fat Coal	4	4000000.00	60002	Fat Coal No.5	9201
2	980002	Washed fat coal	5	5000000.00	60002	Fat Coal No.5	9202
3	980004	Washed gas coal	4	2000000.00	60003	Gas Coal No.3	9203
4	980005	Washed forge coal	5	1000000.00	60004	Forge Coal No.3	9201

QUERY 4

	CoalTransportatio...	CT_Transportation_Origin_Warehou...	CT_Transportation_Target_Warehou...	Transportation_Target_Warehouse_StreetN...	Transportation_Target_
1	5550003	58003	70005	Bernard	Oxnard
2	5550004	58004	70002	Fuda	Cookeville

QUERY 5

	CoalTransportatio...	CT_Transportation_Origin_Warehou...	CT_Transportation_Target_Warehou...	Transportation_Target_Warehouse_StreetN...	Transportation_Target_
1	5550001	58001	70001	Archensha	LittleRock
2	5550002	58002	70001	Archensha	LittleRock

QUERY 6

	Manager...	Manager_JobType	Manager_Department	Manager_FirstNa...	Manager_MiddleNa...	Manager_LastNa...	Manager_...	Manager_Birthday	Ma
1	3001	Prodcuton Managing	Coal mine production dept.	David		Brown	1	1973-04-20 00:00:00.000	12
2	3002	Refinery managing	Refinery factory	Michelle		Davidson	1	1976-07-20 00:00:00.000	9

QUERY 7

	CoalSale_Orde...	CS_CoalBuyerID	CoalSale_Quan...	CoalSale_QualityRequ...	CoalSale_Deliver_Time	CS_Manager...	CS_Transportation_Target_WareHous...
1	20002	1004	6000000.00	3	2008-03-29 00:00:00.000	3005	70002
2	20004	1005	10000000.00	4	2008-04-11 00:00:00.000	3005	70005

QUERY 8

	CoalSale_Orde...	CS_CoalBuyerID	CoalSale_Quan...	CoalSale_QualityRequ...	CoalSale_Deliver_Time	CS_Manager...	CS_Transportation_Target_WareHous...
1	20003	1003	5600000.00	5	2008-04-19 00:00:00.000	3005	70004
2	20005	1001	3000000.00	6	2008-05-01 00:00:00.000	3005	70003

QUERY 9

	CoalSale_Orde...	CS_CoalBuyerID	CoalSale_Quan...	CoalSale_QualityRequ...	CoalSale_Deliver_Time	CS_Manager...	CS_Transportation_Target_WareHou...	CoalB
1	20003	1003	5600000.00	5	2008-04-19 00:00:00.000	3005	70004	2

QUERY 10

	CoalBuyerID	CoalBuyer_Name	CoalBuyer_OfficeNum...	CoalBuyer_StreetNa...	CoalBuyer_CityNa...	CoalBuyer_StateNa...	CoalBuyer_CountryNa...	CoalBuyer_...
1	1003	Ceasur Electricity	353	Taning Street	London	London	England	40745
2	1001	Elon Electricity	1133	Gergoe street	New Brunswick	NJ	U.S.A	08901

QUERY 11

	CoalMine_ID	CoalMine_Pla...	CoalMine_St...	CoalMine_Coun...	CoalMine_Reserv...	CoalMine_EmployeeNum...	CoalBuyer_Name
1	905	Columbus	IN	U.S.A	900000000.00	500	Ceasur Electricity

QUERY 12

	Average a...	Oldest a...	Smallest a...	Average work ye...	Longest work ye...	Shortest work ye...
1	30	35	27	8	12	7

QUERY 13

	Average tranportation ti...	Maximum Transportation ti...	Minimum transportation ti...
1	9	12	8

QUERY 14

	Manager...	Manager_JobType	Manager_Department	Manager_FirstNa...	Manager_MiddleNa...	Manager_LastNa...	Manager_...	Manager_Birthday	Ma
1	3001	Prodcuton Managing	Coal mine production dept.	David		Brown	1	1973-04-20 00:00:00.000	12